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Knowledge-data driven digital twin platform: intelligent prediction and control of tunnel face stability during large-diameter slurry shield construction

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ABSTRACT

Large diameter shield construction (LDSS) safety is particularly important. To ensure safe excavation in LDSS projects, this paper proposed a digital twin (DT) platform integrated with knowledge-data driven method to achieve prediction and control of tunnel face stability (TFS) during LDSS construction. The DT platform enables acquisition of physical data and expert knowledge, facilitating bidirectional synchronization between physical entities and virtual counterparts. DT platform utilizes Bayesian Optimization (BO), Graph Convolutional Network (GCN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and SHapley Additive exPlanations (SHAP), driven by physical data, together with expert knowledge, to achieve knowledge-data driven prediction and control of Tunnel Face Stability (TFS) during LDSS construction. A case study of Wuhan Metro Line 12 construction demonstrates the platform's effectiveness, with key findings revealing: (1) The DT platform achieves accurate TFS prediction across six Geological types, showing average R^2 values of 0.935 and root mean square error (RMSE) of 0.239. (2) Key construction parameters are identified through the knowledge-data driven method, including air chamber pressure, grouting pressure, advance rate, cutterhead rotation speed, and cutterhead torque. (3) The DT platform enables control of the TFS during LDSS construction by maintaining the slurry pressure (SP) within an optimal range. The DT platform developed in this study fills TFS research gaps in LDSS construction and provides valuable references for similar complex projects.

1. Introduction

With the rapid advancement of tunneling engineering technology, large-diameter slurry shield (LDSS) technology has emerged and gained widespread adoption (Chen et al., 2024, 2025; Wu et al., 2023). it has been applied in significant projects such as the Slender West Lake Tunnel in Yangzhou (Chenn kui, 2014) and the Sanyang Road Tunnel in Wuhan (Wang, 2015). However, the diversity of the geological environments and the uncertainty of the construction scenarios have significantly increased the safety risks associated with LDSS operations (Wu et al., 2024), particularly the precise prediction and control of tunnel face stability (TFS) (Xu et al., 2021). As the diameter of the shield increases, the geological structures encountered by the tunnel become more complex (Li et al., 2017; Yu et al., 2023), and the frequency of encountering multilayered strata and composite strata with varying mechanical properties (Chen et al., 2025), such as intermixed soft and hard clay and sand, significantly increases (Cong et al., 2023). These challenges not only markedly increase the complexity of the excavation environment but also greatly increase the difficulty and intricacy of maintaining TFS. Therefore, the TFS is crucial for ensuring the safe construction of an LDSS.

Traditional research on TFS issues has been mostly centered around theoretical analysis, numerical simulation, and experimental studies. The recent progress in shield machine (SM) monitoring technology, coupled with the deep integration of data-driven methods, is now driving the field of shield tunnel construction towards a more advanced

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and intelligent stage of development (Hu et al., 2022; Li et al., 2023b; Wang and Tian, 2024; Yuan et al., 2025).The various common research methods are summarized in Table 1.

This analysis highlights significant advancements over previous studies in the field but also notes certain limitations:

(1) Theoretical analysis simplifies complex situations for easier understanding and computation, but this may reduce prediction accuracy by omitting some influencing factors.
(2) Numerical simulations need fine meshing and many iterations to meet high accuracy demands, which increases computational costs and makes large-scale simulations time-consuming, affecting their application efficiency in construction.
(3) Experimental studies in complex environments require substantial investment in equipment and manpower, and they interrupt the construction process, leading to higher costs and safety risks.
(4) Traditional data-driven methods can build predictive models using field data but lack interpretability and struggle with highly complex parameter interactions and dynamic changes.

As a branch of data-driven methods, deep learning has been extensively utilized by many scholars owing to its powerful data processing and analysis capabilities, as well as its capacity to reveal and capture complex nonlinear relationships within data (Liu et al., 2021; Phoon and Zhang, 2023; Zhang et al., 2020).。 However, existing research indicates that the construction parameters in SM construction exhibit spatiotemporal relationships; these relationships are difficult for conventional data-driven methods to capture and analyze, and their predictive performances are reduced (Fu et al., 2023; Zhang et al., 2024b). Moreover, some existing studies mainly focus on the accurate prediction of TFS under traditional SM conditions; however, there is limited research on establishing suitable data-driven methods for precise prediction and control of TFS under LDSS conditions.

Table 1
Summary and comparison of several studies on TFS issues.

Method	Reference	Advantages	Assumption
Theoretical analysis	Man et al. (Man et al., 2022) Shi et al. (Shi et al., 2025a) Man et al. (Man et al., 2024)	Provides an in-depth understanding of the fundamental principles and concepts governing tunnel face stability	The material properties and boundary conditions are assumed to be known and constant.
Numerical simulation	Li et al. (Li et al., 2023a) Yang et al. (Yang et al., 2024) Tu et al. (Tu et al., 2024)	Able to simulate intricate physical phenomena and engineering challenges, encompassing multiscale and nonlinear aspects.	The meshing and physical parameter settings in the computational model are assumed to be reasonable.
Experimental studies	Lü et al. (Lü et al., 2018) Cui et al. (Cui et al., 2025) Liu et al. (Liu et al., 2024)	Provides direct evidence for validating the theoretical analysis and numerical simulation results.	Assuming the control variables in the experiment can effectively simulate the conditions in actual engineering.
Data-driven method	Liu et al. (Liu et al., 2021) Zhao et al.(Zhao et al., 2023) Wang et al. (Wang et al., 2021a)	Capable of processing and analyzing large datasets and reducing research costs and risks.	Assuming there is a sufficient amount of high-quality data to train and validate the model.

Among the various data-driven methods, LSTM is commonly used for long time series data but can only predict future moments based on past information (Li et al., 2019). This is insufficient for LDSS construction parameters, which may also depend on future moments (Chen et al., 2022; Siami-Namini et al., 2019). To address this, BiLSTM, combining forward and backward LSTMs, is introduced to process LDSS tunnel construction data (Graves and Schmidhuber, 2005). Additionally, TFS in LDSS construction is influenced by both past construction parameters and spatial relationships among parameters (Fu et al., 2024). For example, different parameters are emphasized in soft soil and hard rock (Chen et al., 2022; Zhao et al., 2019). To capture these spatial relationships, GCNs are introduced, representing data as graph structures and using the Laplacian matrix for convolution operations.

During the development of data-driven method (Wu et al., 2026), hyperparameter optimization is a critical preliminary step aimed at identifying the hyperparameter configurations that achieve optimal model performance during both the training and testing phases (Xu et al., 2025). While traditional optimization methods, such as manual operation (MO), grid search (GS), and random search (RS), provide researchers flexibility in model tuning, they suffer from inefficiency, subjective bias, wasted computational resources, and inadequate optimization (Victoria and Maragatham, 2021). In contrast, Bayesian optimization (BO) introduces a probabilistic acquisition function to efficiently explore the parameter space which significantly increases the likelihood of discovering the global optimum (Chen et al., 2025). Therefore, to improve the prediction and control performance of data-driven methods, a hybrid BO-GCN-BiLSTM method is proposed.

Additionally, to address the research gap in key construction parameters for TFS during LDSS construction, this paper employs the SHapley Additive exPlanations (SHAP) to identify and analyze these key parameters. By utilizing this method, it is possible to accurately determine the construction parameters that significantly impact TFS (Xu et al., 2026), thereby enabling precise control of TFS in LDSS construction.

Data-driven methods can improve prediction and control performance but may lack reliability for practical engineering due to the absence of practical experience and expert knowledge (Zhang et al., 2024a). Therefore, a knowledge-data driven method integrating expert knowledge is more suitable for guiding TFS during LDSS construction. This approach can select input parameters and identify key influencing factors based on expert knowledge and literature (Su et al., 2025). However, most existing research focuses on developing prediction and control algorithms, with limited efforts on creating a unified platform for integrating knowledge-driven and data-driven methods with physical systems to enhance LDSS control and monitoring capabilities.

Digital twin (DT) technology integrates construction site data with virtual dynamic models to create a digital simulation platform, showing superior capabilities in tunnel operation and management (Boje et al., 2020; Yu et al., 2021, 2023). Recent studies have applied the concept of DT to general shield tunnel excavation, enabling efficient remote interaction between physical and virtual entities by transforming physical equipment into digital models (Pan and Zhang, 2021; Zhang et al., 2024b). However, research on using DT platforms for data collection and TFS prediction and control in LDSS construction is limited. This paper proposes a DT platform that integrates data-driven methods (BO-GCN-BiLSTM and SHAP) with expert knowledge to form a knowledge-data driven method, enabling data acquisition and supporting TFS prediction and control in LDSS construction.

In summary, the following issues motivate this paper: (1) how to analyze the spatiotemporal relationships among the various construction parameters affecting the TFS during LDSS construction; (2) what construction parameters are the key factors affecting the TFS during LDSS construction; and (3) how to implement prediction and control of TFS in LDSS construction. To address these issues, a knowledge-data driven method is developed for predicting and controlling TFS in LDSS

construction. A DT model for LDSS construction is also proposed to enable TFS prediction and control. The primary innovations can be summarized as follows:

(1) A novel BO-GCN-BiLSTM method is pioneered and developed to capture the spatiotemporal relationship of the construction parameters affecting TFS during LDSS construction. By harnessing the strengths of BO, GCN, and BiLSTM, this method provides an advanced framework for prediction and control of the TFS of LDSS.
(2) Utilizing the SHAP and expert knowledge, the key construction parameters influencing TFS during LDSS construction can be identified. Its application, by elucidating these key construction parameters, provides a foundation for achieving effective control of TFS during LDSS construction and enhances the interpretability of BO-GCN-BiLSTM model.
(3) A DT platform is proposed for achieving data acquisition, prediction, and control of TFS in LDSS construction. The DT platform employs a knowledge-data driven approach by integrating expert knowledge with BO-GCN-BiLSTM and SHAP, providing guidance for more accurate and reasonable prediction and control of TFS.

The remainder of this paper is organized as follows. Section 2 details the proposed methodology. Section 3 showcases the proposed DT platform's effectiveness through a real engineering project. Section 4 provides an in-depth discussion of the paper. Section 5 summarizes the conclusions of this paper and outlines future research directions.

2. Methodology

To enable prediction and control of the TFS during LDSS construction, a DT platform has been developed. This platform includes a virtual twin layer with a virtual LDSS model, a data-driven module, and a reality service layer. Specifically, the data-driven module integrates BO, GCN, and BiLSTM to capture the spatiotemporal relationships of shield construction parameters. Additionally, the reality service layer facilitates prediction, SHAP for key construction parameters, and control of TFS during LDSS construction using the knowledge-data driven method. Fig. 1 illustrates the framework of the DT platform.

2.1. Virtual twin layer

The establishment of the virtual twin layer is a core component of the DT platform. It achieves digital mapping and real-time interaction with the LDSS physical entity through high-fidelity modeling of the LDSS virtual model and multidimensional construction data fusion. This layer includes a data acquisition module and a virtual model.

During the LDSS tunnel construction process, the digital twin (DT) platform utilizes two types of data sources: physical data and expert knowledge (Li et al., 2025). The former consists of real-time and historical data collected from the construction site and the shield machine, including ring numbers, cutter head speed, cutter head torque, and tunneling speed; the latter primarily relies on experts who, based on their expert knowledge and relevant literature (Chen et al., 2024), guide the selection of key parameters, thereby enabling knowledge-data-driven methods for TFS prediction and control during LDSS construction. These two data categories jointly support monitoring, prediction, and control functionalities.

However, they can only provide a static description of the construction conditions and are unable to reflect the dynamic evolution of the LDSS–tunnel interaction in real time. To address this limitation, the platform establishes a virtual model based on the physical entities, fully digitizing the LDSS and tunnel system, and continuously updates it with real-time data. Interaction between the physical and virtual systems can be enabled through signal transmission via the Internet of Things (IoT) and 5th generation of cellular networks (5G) networks (Zhang et al., 2024b). Field data from the physical LDSS are uploaded via signals to the project manager's online DT platform. The platform then performs key parameter prediction and control based on the virtual LDSS model, and the resulting instructions are transmitted back through signals to the physical LDSS on-site, guiding the operator's actions (He et al., 2024). This bidirectional data flow establishes a closed-loop interaction between the physical entity and its virtual counterpart, as illustrated in Fig. 2.

2.2. Data-driven module

2.2.1. Physical data preprocession

With physical data collected from LDSS, the DT platform preprocesses this data and can integrate data-driven module to provide,

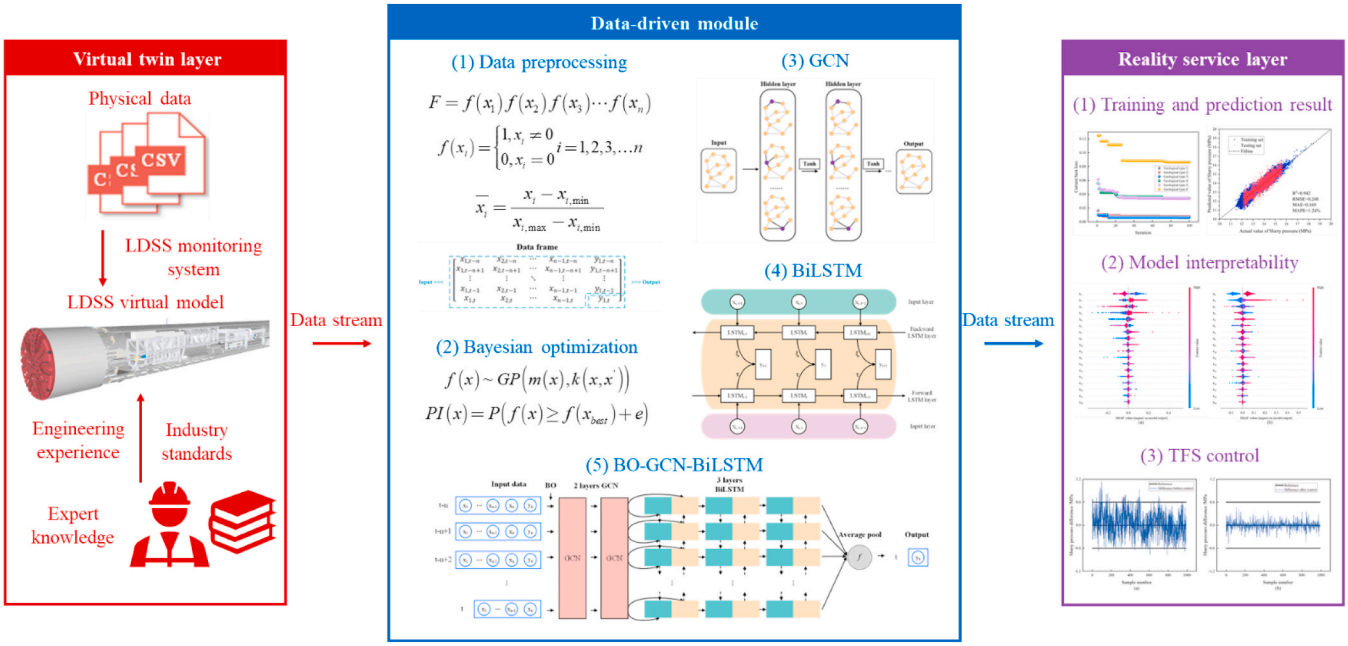


Fig. 1. Framework of TFS prediction based on DT platform.

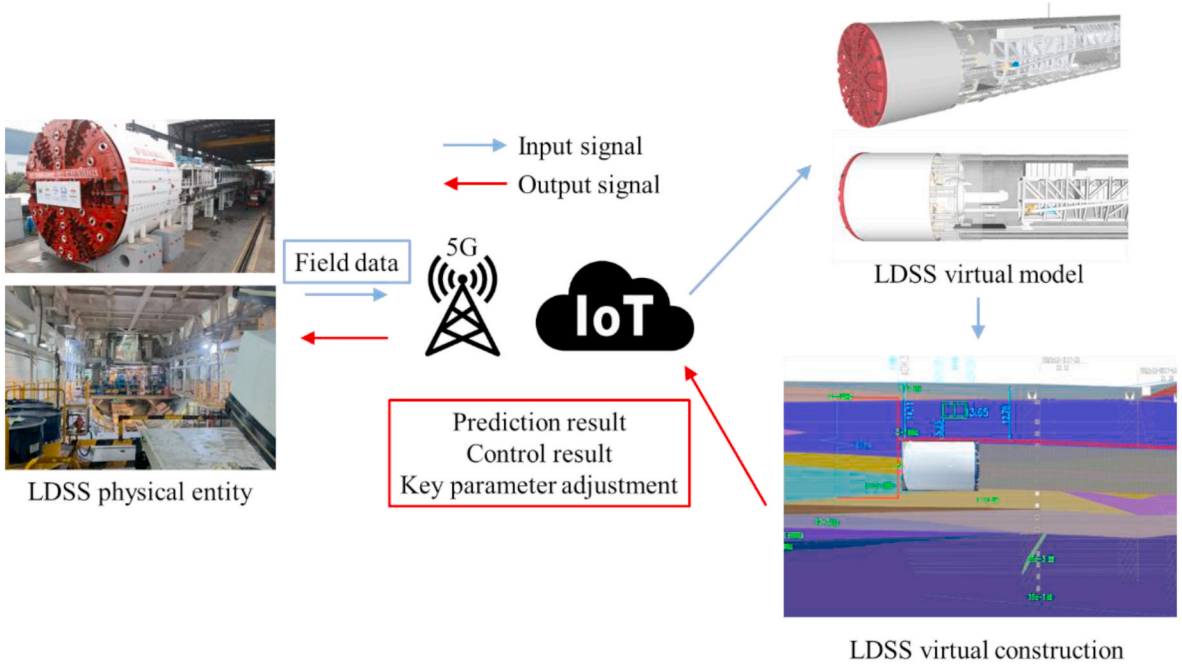


Fig. 2. Interaction between LDSS physical entities and virtual models.

reliable prediction and control of TFS in LDSS construction. Generally, data preprocessing includes deleting zero data and performing data normalization. A row of LDSS construction parameter data is defined as $X_i = \{x_1, x_2, x_3, \ldots, x_n\}$, where $x_1, x_2, x_3, \ldots, x_n$ represent different parameters. The processes of removing null data and zero data and performing data normalization are represented by Eqs. (1)–(3) (Chen et al., 2023; Feng et al., 2021; Wang et al., 2023):

$$F = f(x_1)f(x_2)f(x_3)\cdots f(x_n) \tag{1}$$

$$f(x_i) = \begin{cases} 1, x_i \neq 0 \\ 0, x_i = 0 \end{cases} i = 1, 2, 3, \ldots n \tag{2}$$

$$\overline{x_i} = \frac{x_i - \min\limits_{1\le i\le n}\{x_i\}}{\max\limits_{1\le i\le n}\{x_i\} - \min\limits_{1\le i\le n}\{x_i\}} \tag{3}$$

where $f(x_i)$ is a judgment function concerning x_i, F is the product function of $f(x_i)$ values, $\overline{x_i}$ is normalized by x_i, $\max\limits_{1\le i\le n}\{x_i\}$ and $\min\limits_{1\le i\le n}\{x_i\}$ represent the maximum and minimum values of x_i.

To better capture the spatiotemporal relationships among the construction parameters, after performing the aforementioned processing, the data are restructured into a matrix that describes the time steps. In this matrix, the row $\{x_{1,t-n}, x_{2,t-n}, \ldots, x_{n-1,t-n}, x_{n,t-n}\}$ indicates the numerical value of each parameter at time step $t-n$, and the column $\{x_{1,t-n}, x_{1,t-n+1}, \ldots, x_{1,t-1}, x_{1,t}\}^T$ indicates the value of parameter x_1 across all time steps. Fig. 3 shows a schematic diagram of the matrix after data restructuring.

2.2.2. LDSS construction parameter selection

In the knowledge-data-driven method of the DT platform, the identification of input and output parameters is crucial for achieving reliable TFS prediction and control during LDSS construction. These parameters are determined collaboratively by experts based on their knowledge, LDSS operators, and project managers within the DT platform, with the parameter selection interface shown Fig. 4. Notably, an intrinsic relationship exists between slurry pressure (SP) and TFS during LDSS operation (Hou et al., 2022). The slurry serves not only to transport excavated soil but also acts as a pressure medium to balance the lateral earth and water pressures on the excavation face (Wu et al., 2024). Additionally, it forms a protective filter cake that helps prevent face instability and collapse. Therefore, SP is established as the key output target in the TFS prediction model.

LDSS construction parameters are transmitted to the virtual twin layer via CSV data files from the LDSS monitoring system, containing various construction parameters. In this paper, the representative construction parameters are selected based on previous research (Cao et al., 2023; Wang et al., 2021b; Zhonghai et al., 2019; Zhou et al., 2023) and expert knowledge from virtual twin layer. Construction parameters chosen as inputs for the data-driven method include the air cushion chamber pressure (x_1), inflow slurry density (x_2), slurry flow (x_3), discharge slurry density (x_4), discharge slurry flow (x_5), in and out of the slurry pressure difference (x_6), slurry injection pressure (x_7), grout

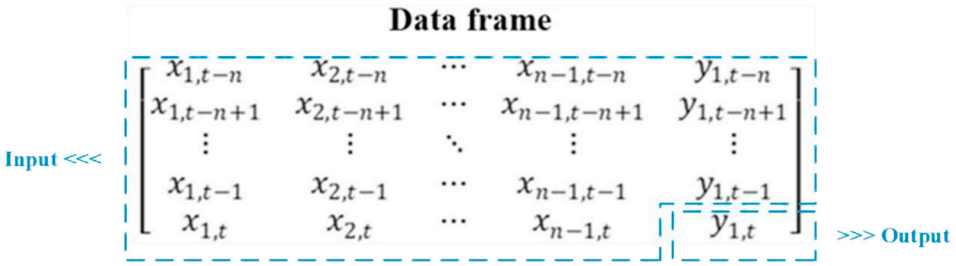


Fig. 3. Schematic diagram of data reconstruction.

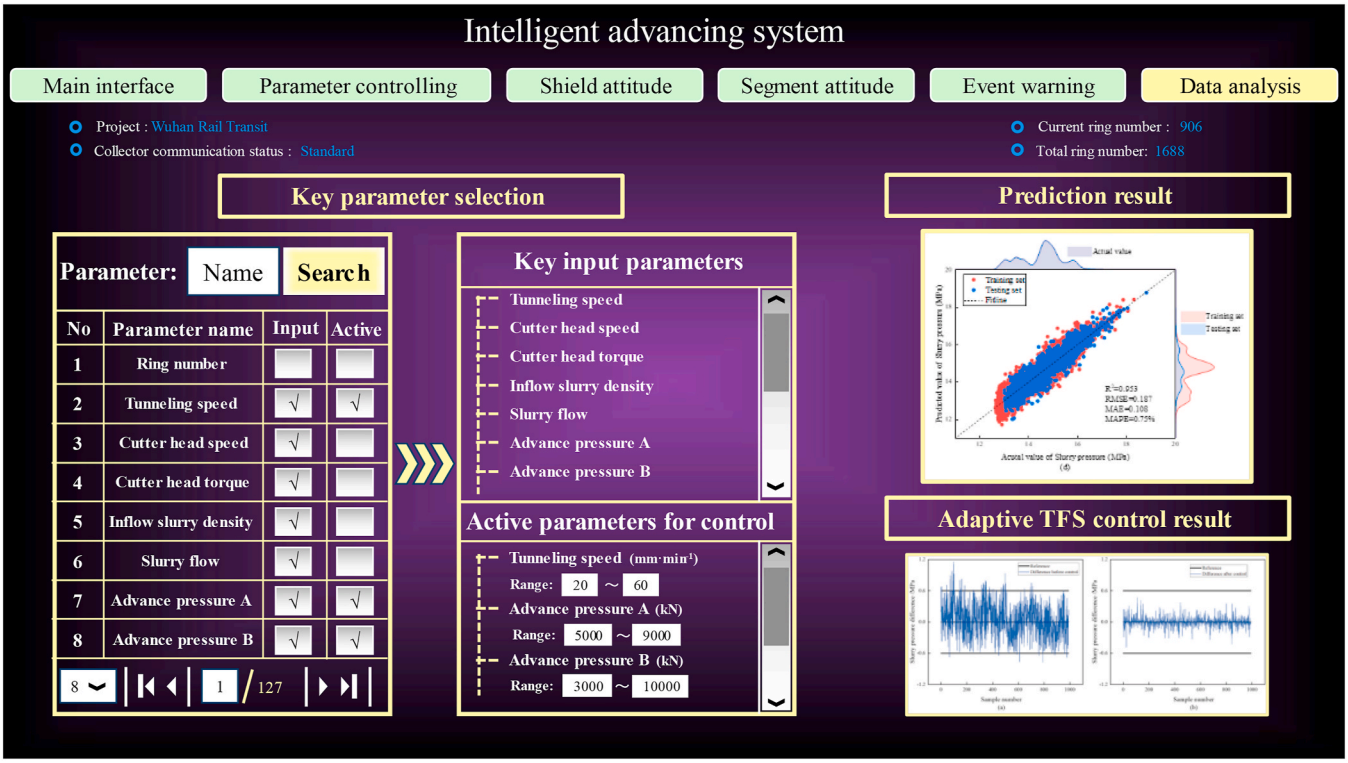


Fig. 4. DT platform parameter selection interface demonstration.

amount (x_8), tunneling speed (x_9), cutter head speed (x_{10}), cutter head torque (x_{11}), cutter extrusion pressure (x_{12}), advance pressure A-E (x_{13}-x_{17}), and penetration (x_{18}). Above all, LDSS construction parameters are listed in Table 2.

2.2.3. BO-GCN-BiLSTM construction

2.2.3.1. Bayesian optimization algorithm. BO is a sequential strategy for hyperparameter search that employs a probabilistic model to guide the search process (Cao et al., 2024), primarily consisting of two essential elements: a probabilistic surrogate model and an acquisition function for guiding parameter selection (Lei et al., 2021). As discussed in the introduction, traditional hyperparameter tuning methods often lack flexibility and suffer from high computational costs (Liu et al., 2024), limitations that become more pronounced when applied to complex data-driven methods. Furthermore, given the diverse geological conditions encountered during LDSS construction, the hyperparameters of data-driven models require frequent adaptation (Liu et al., 2023). Therefore, integrating Bayesian optimization is essential to enhance both the predictive performance and computational efficiency of data-driven approaches (Xu et al., 2025). Gaussian process regression (GPR) is the most widely adopted probabilistic model in BO due to its high flexibility and strong generalization capability. The GPR model is expressed in Eq. (4) (Mahmoodzadeh et al., 2022; Yang et al., 2023):

$$f(x) \sim GP(m(x), k(x, x')) \tag{4}$$

where $m(x)$ denotes the mean value function and $k(x, x')$ represents the covariance function. This formula allows predictions to be made regarding new input points x and provides uncertainty estimates for the predictions, given the training data (Chen et al., 2023).

Table 2
Detailed information on the LDSS construction parameters.

Type	Construction parameter	Type	Construction parameter
Input	Air cushion chamber pressure (x_1), MPa	Input	Cutter head torque (x_{11}), kN·m
	Inflow slurry density (x_2), kg/m^3		Cutter extrusion pressure (x_{12}), kN
	Slurry flow (x_3), m^3/h		Advance pressure A (x_{13}), kN
	Discharge slurry density (x_4), kg/m^3		Advance pressure B (x_{14}), kN
	Discharge slurry flow (x_5), m^3/h		Advance pressure C (x_{15}), kN
	In and out of the slurry pressure difference (x_6), MPa		Advance pressure D (x_{16}), kN
	Slurry injection pressure (x_7), MPa		Advance pressure E (x_{17}), kN
	Grout amount (x_8), m^3		Penetration (x_{18}), mm
	Tunneling speed (x_9), mm/min	Output	Slurry pressure (y_1), MPa
	Cutter head speed (x_{10}), rpm		

2.2.3.2. GCN. During the LDSS construction, TFS is influenced by the spatial relationships between construction parameters. To capture the spatial relationships of construction parameters under varying geological conditions, this paper employs GCN to extract these relationships. In a graph $\Gamma = (\varsigma, E)$, ς is the set of nodes, E is the set of edges, and each node i has features χ_i, which are denoted by X. X is an $N \times F$-dimensional matrix, where N denotes the number of nodes and F denotes the number of features or dimensions of each node. An adjacency matrix A also exists for the graph Γ. The output dimension is D. The propagation processes between the layers of the GCN are shown in Eqs. (5)–(7) (Ghayekhloo and Nickabadi, 2023):

$$\widehat{A}' = D^{\left(-\frac{1}{2}\right)} \widehat{A} D^{\left(-\frac{1}{2}\right)} \tag{5}$$

$$X' = XW \tag{6}$$

$$H = \sigma(\widehat{A}' X') \tag{7}$$

where σ denotes the activation function, ReLU; D is degree matrix; $\widehat{A}$ is

adjacency matrix containing information about its own nodes; $\widehat{A} = A + I$, with I as the identity matrix; $\widehat{A}'$ is obtained by normalizing $\widehat{A}$; X′ is the matrix obtained after weight transformation of X; and H represents the learned node embeddings.

Based on the research of other scholars (Fu et al., 2024; Pan et al., 2022), the structure of the GCN is shown in Fig. 5.

2.2.3.3. BiLSTM. The construction parameters of LDSS at the current time are not only related to historical data but may also be associated with future data. To address this characteristic, this paper employs a Bidirectional Long Short-Term Memory (BiLSTM) network to capture the temporal relationships among construction parameters. The BiLSTM model extends the traditional LSTM architecture by processing sequences in both forward and backward temporal directions, thereby capturing comprehensive contextual dependencies (Tornyeviadzi et al., 2023). This allows it to effectively encode the dependencies between LDSS construction data and the target parameters. The outputs of the BiLSTM model are shown in Eqs. (8) and (9) (Lei et al., 2024):

$$\begin{cases} \overrightarrow{h_t} = \overrightarrow{LSTM}\left(\overrightarrow{h_{t-1}}, x_{i,t}\right), t \in [1, T] \\ \overleftarrow{h_t} = \overleftarrow{LSTM}\left(\overleftarrow{h_{t-1}}, x_{i,t}\right), t \in [1, T] \end{cases} \tag{8}$$

$$y_t = \tau \overrightarrow{h_t} + \xi \overleftarrow{h_t} \tag{9}$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ are the target parameter outputs of the $\overrightarrow{LSTM}$ and the $\overleftarrow{LSTM}$ at time t, respectively. $\overrightarrow{h_{t-1}}$ and $\overleftarrow{h_{t-1}}$ are the target parameter outputs of the $\overrightarrow{LSTM}$ and the $\overleftarrow{LSTM}$, respectively, at time $t-1$, and $x_{i,t}$ is the model input parameter x_i at time t. y_t is the target parameter prediction of BiLSTM. The values of the τ and ξ are obtained by training on the LDSS construction data. T is the length of time series. The typical structure of BiLSTM is shown in Fig. 6.

2.2.3.4. BO-GCN-BiLSTM. This paper propose a BO-GCN-BiLSTM method that integrates BO, GCN, and BiLSTM networks, with the aim to optimize the accuracy of TFS prediction; in particular, in parsing the spatiotemporal relationship among the LDSS construction parameters, our method demonstrates superior performance. In this study, a two-layer GCN is configured with 64 and 32 neurons in the first and second convolutional layers, respectively, using the Tanh activation function. A three-layer BiLSTM is employed with 128, 64, and 32 bidirectional neurons in each layer, respectively. The sequence length is determined based on the length of the time series collected under different geological conditions, and the Tanh activation function is also applied. The initial learning rate for both models is set to 0.001, serving as input for the BO process. Fig. 7 shows the construction logic of the BO-GCN-BiLSTM method.

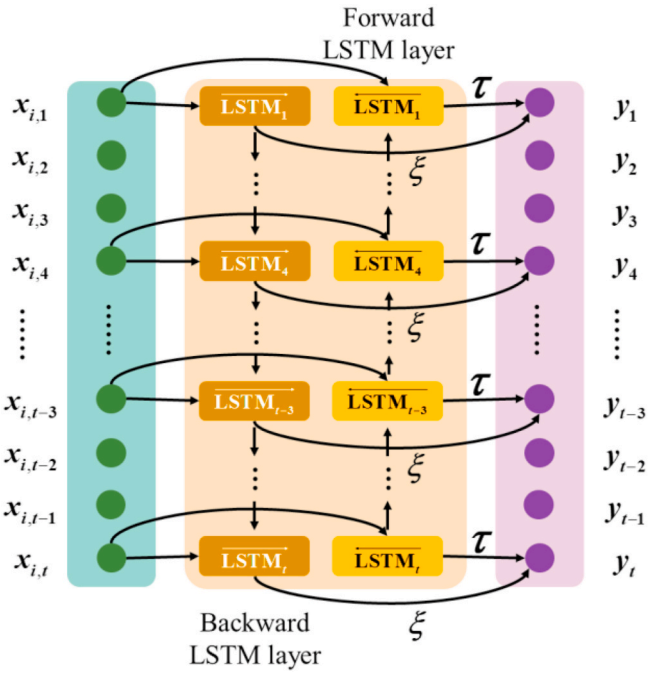


Fig. 6. Typical structure of BiLSTM.

The specific pseudocode of the BO-GCN-BiLSTM method is shown in Algorithm 1.

Algorithm 1. BO-GCN-BiLSTM pseudocode

Input: Graph data $G = (V,E)$, Node features X, Time series data Y, Number of epochs EPOCHS, Hyperparameter search space PARAM_SPACE
Output: Predicted time series Z
1: **procedure BO-GCN_BiLSTM_PREDICTION**(G, X, Y, PARAM_SPACE)
2: BEST_PARAMS = None

(*continued on next page*)

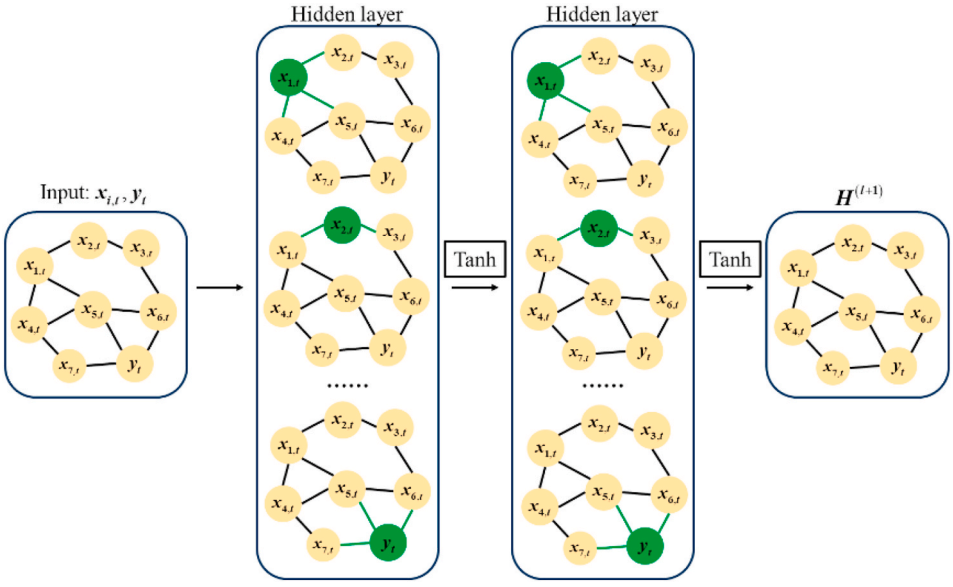


Fig. 5. Typical structure of GCN.

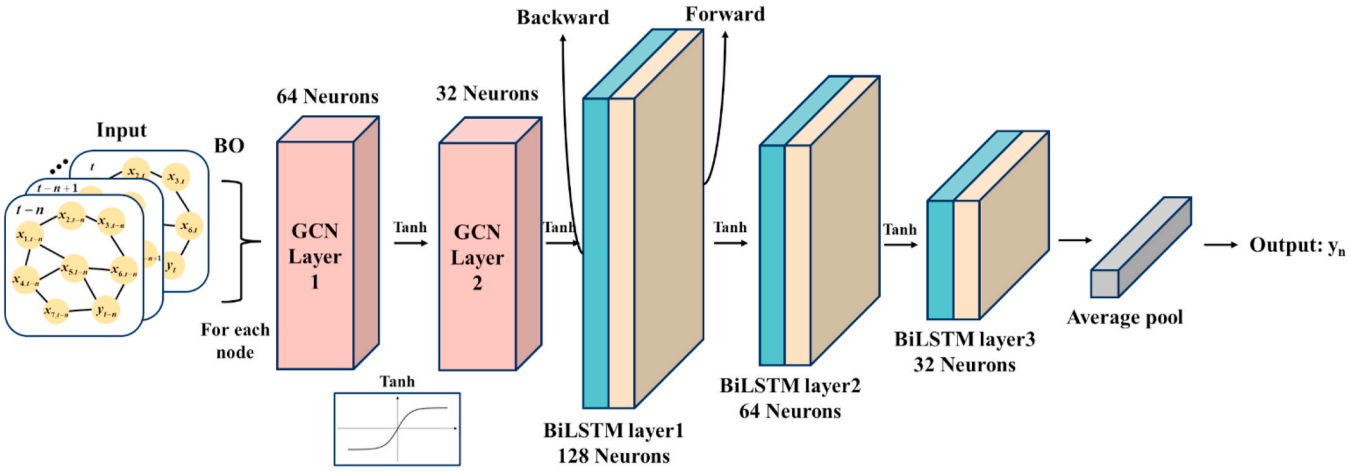


Fig. 7. Schematic diagram of BO-GCN-BiLSTM.

(*continued*)

```
BEST_LOSS = ∞
while not convergence do
    HP = BayesianOptimize(PARAM_SPACE)
    if train(G, X, Y, HP) improves performance then
        BEST_PARAMS = HP
        BEST_LOSS = current_loss
    end if
end while
return BEST_PARAMS, BEST_LOSS
end procedure
procedure BayesianOptimize(PARAM_SPACE)
while not termination do
    next_HP = SelectNextHP(PARAM_SPACE)
    evaluatePerformance(next_HP)
    update(PARAM_SPACE; next_HP; performance)
end while
end procedure
procedure train(G, X, Y, HP)
Initialize GCN-BiLSTM with HP
for each epoch do
    Forward and Backward propagation on G, X, Y
    Update model parameters
end for 26: return model_performance
end procedure
```

2.3. Reality service layer

The established data-driven module within the DT platform can be further extended to provide reliable services for LDSS tunnel excavation (Cao et al., 2025). The reality service layer is a crucial component of the DT platform, with its core function being the integration of the virtual simulation capabilities of the virtual twin layer with the actual operational requirements of the LDSS physical entity. This enables monitoring of LDSS construction and prediction and control of TFS.

In the context of prediction and control of TFS during LDSS construction, the reality service layer integrates physical data and expert knowledge to determine the input and output parameters for TFS prediction. It constructs a knowledge-data driven method based on BO, GCN, and BiLSTM. This method leverages BO to optimize hyperparameters, thereby enhancing prediction accuracy and generalization capability. Simultaneously, it employs SHAP and expert knowledge to select critical parameters, ensuring that the model accurately identifies key construction parameters influencing TFS.

This process, based on the knowledge-data driven method, not only achieves prediction and control of TFS but also provides scientific guidance for adjusting the construction parameters of LDSS through feedback mechanisms. The training and evaluation of the BO-GCN-BiLSTM model, as well as the control flow for TFS, are described below.

2.3.1. BO-GCN-BiLSTM training and evaluation

The preprocessed data are used to train the BO-GCN-BiLSTM method. The 80% of the dataset allocated for BO-GCN-BiLSTM training and 20% allocated for BO-GCN-BiLSTM testing. The training process begins with X_{train}; after a certain number of iterations, the Y_{train} is obtained. Mean squared error (MSE) is the loss function (Liu et al., 2024). After training, X_{test} is tested on the test set data. After a certain number of iterations, the corresponding Y_{test} is obtained and compared with the actual values (Wu et al., 2023).

To evaluate the prediction performance, the mean absolute error (MAE) (Liu and Chen, 2025), root mean squared error (RMSE) (Luo et al., 2025), mean absolute percentage error (MAPE) (Liu et al., 2025), and R^2 are selected as the evaluation indices (Liu et al., 2020). The definitions of these four indices are provided in Eq. (10) to Eq. (13) (Guo et al., 2023; Pan et al., 2022):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|f(x_i) - ob\{y_i\}\right| \tag{10}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[f(x_i) - ob\{y_i\}\right]^2} \tag{11}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{f(x_i) - ob\{y_i\}}{f(x_i)}\right| \tag{12}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(ob\{y_i\} - f(x_i)\right)^2}{\sum_{i=1}^{n}\left(ob\{y_i\} - mean\{y_i\}\right)^2} \tag{13}$$

where n is the data volume, $ob\{y_i\}$ is the observed value (Liu et al., 2025), $f(x_i)$ is the predicted value, and $mean\{y_i\}$ is the mean of the observed values.

2.3.2. Model interpretability based on SHAP

SHAP, inspired by game theory, innovatively attributes a model's prediction output to the individual contribution of each input feature and quantifies the importance of each feature (Ma et al., 2020). In the context of LDSS excavation, this can guide operators in selecting key construction parameters that influence target variables, while enhancing the interpretability of data-driven methods and providing more reliable model explanations. Specifically, SHAP defines the "contribution" of each input feature to the prediction outcome as the marginal payoff in a cooperative game, and fairly allocates the total payoff using Shapley values. This additive explanatory model is expressed as Eq. (14) (Fu et al., 2025):

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z_i' \tag{14}$$

where $g(z')$ denotes the additive explanatory model; $z_i' \in (0,1)$ is the binary simplified input vector, with each component z' indicating the presence $z_i' = 1$ or absence $z_i' = 0$ of the feature i; ϕ_0 represents the baseline value; ϕ_i is the Shapley value of the feature i, quantifying its contribution to the prediction; M is the total number of input features.

To ensure a unique solution for the equation, the additive explanatory model must satisfy the following conditions: (1) Local Accuracy: the model output must exactly match the original model's prediction for the target input x and its simplified representation z'; (2) Missingness: features that are masked $z_i' = 0$ should have no contribution to the output; (3) Consistency: if a feature's marginal contribution to the model prediction increases after a change in the model, its Shapley value must also increase. When these three conditions are satisfied, the solution to the equation is uniquely given by Eq. (15) (Li et al., 2024):

$$\phi_i(f,x) = \sum_{S \subseteq \{1,2,\ldots,M\} \setminus \{i\}} \frac{|S|!(M-|S|-1)!}{M!} [f(x_{S\cup\{i\}}) - f(x_S)] \tag{15}$$

where $\phi_i(f,x)$ represents the Shapley value of the feature i; f represents the data-driven method; x represents the complete input sample that needs to be explained; S represents a feature subset that does not include feature i, $S \subseteq \{1,2,\ldots,m\}$; $|S|$ represents the number of elements in the subset S; $x_{S\cup\{i\}}$ represents the value obtained by retaining only the subset S and the value of feature i based on the original input x; x_S represents the original input x with only the values of subset S retained, excluding feature i.

2.3.3. TFS control strategy

To accurately simulate the control process of TFS, this paper introduces a knowledge-data driven method based control strategy in the DT platform, where the latest control iteration results replace historical data, ensuring each iteration is based on the outcomes of the previous regulation. The main steps are as follows:

a. Determine Control Objectives: By consulting relevant literature and expert knowledge within the virtual twin layer, and integrating practical construction requirements and standards, establish the reasonable range for control objectives in actual construction.
b. Identify Key Parameters: Experts utilize their own expert knowledge within the virtual twin layer and SHAP, determine the key construction parameters and their adjustment ranges, which serve as input parameters for TFS control.
c. Control Iteration: Utilize physical data from the virtual twin layer and the data-driven module to perform control iterations, with each iteration result updating and replacing historical data.
d. Stop Condition: When the control objectives fall within the predefined reasonable range, the iteration process stops.

This knowledge-data driven method not only enhances the timeliness and adaptability of the prediction model but also provides a solid foundation for continuous control of LDSS construction safety and excavation efficiency. The specific control flow is illustrated in Fig. 8.

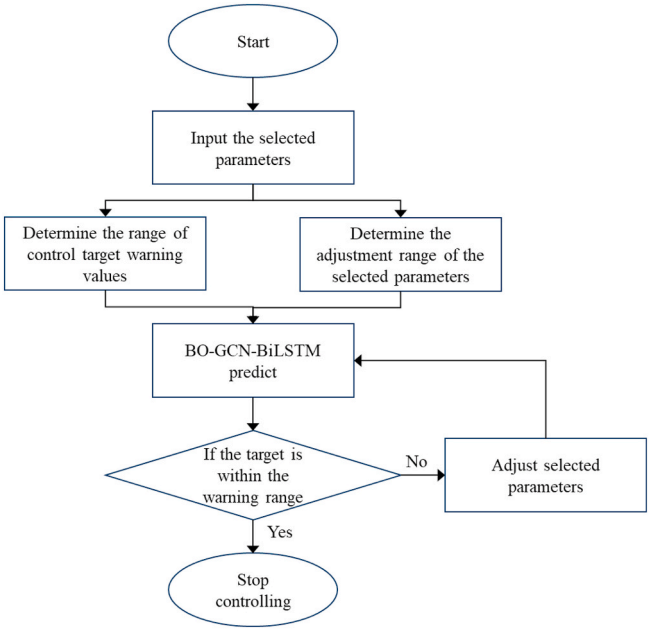


Fig. 8. DT platform controlling process of TFS.

3. Case study

3.1. Case background and DT platform interface

This paper is based on a section of the longest metro loop in Asia - Wuhan Metro Line 12 between Lingwucun Station and Guobo Center South Station; in this section, the Yangtze River tunnel segment is 2610 m long and is excavated by two Φ12.56-m LDSS machines. The tunnel is designed with a single-bore double-track layout and features an outer segment diameter of 12.1 m, an inner diameter of 11.1 m, a segment thickness of 0.5 m, and a ring width of 2 m. During shield advancement, all relevant construction parameters are recorded by the shield data acquisition system. Fig. 9(a and b,c,d) shows the Location Map of Wuhan Metro Project, Construction site photos, Interior view of the tunnel, and Data acquisition, respectively. Fig. 10 shows the geological conditions for LDSS construction. And Fig. 11 shows the DT platform interface of LDSS construction.

3.2. Data collection

To address the inadequacies of traditional methods in the TFS during LDSS construction research and to achieve effective TFS prediction and control, the proposed knowledge-data driven method requires preprocessing of the collected data before training. The preprocessing steps include data cleaning, normalization, and reconstruction (Chen et al., 2025). The purpose of this process is to remove invalid and outlier data from the raw data. Normalization is applied to eliminate the impact of outlier sample data on model training. Finally, the data are reconstructed to prepare them for model input. All data used in this paper were obtained from the monitoring system, primarily comprising construction parameters related to rings 30 to 471. These data encompass complex and diverse geological conditions, which can be broadly categorized into six geological types. In this section, Geological type 1 to Geological type 6 are selected for TFS prediction. The property parameters of each Geological type as shown in Table 3. The construction parameters for each Geological type are described in Table 4. The division of datasets for each geological type is illustrated in Fig. 12.

3.3. Knowledge-data driven method based TFS prediction

Based on the reality service layer of the DT platform, the LDSS construction process utilizes the knowledge-data driven method, including BO-GCN-BiLSTM method, physical data and expert knowledge. First, BO is used for hyperparameter tuning. Then, time steps for the BO-GCN-BiLSTM method are set, with each time step being predicted individually. The optimal time step is selected based on a combination of prediction accuracy and operational efficiency.

3.3.1. Bayesian hyperparameter optimization

BO-GCN-BiLSTM model have complex structures, and their

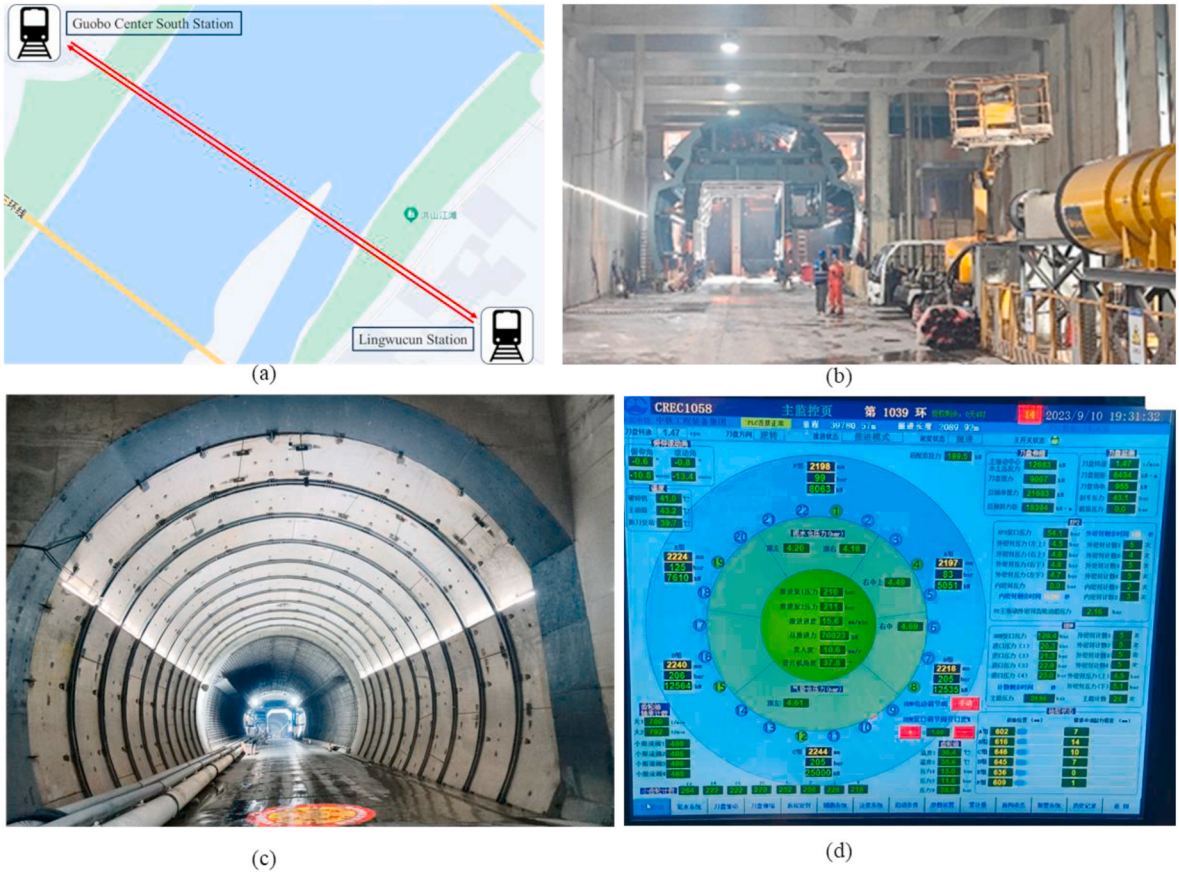


Fig. 9. Project information for the (a) Location Map of Wuhan Metro Project, (b) Construction site photos, (c) Interior view of the tunnel (d) Data acquisition.

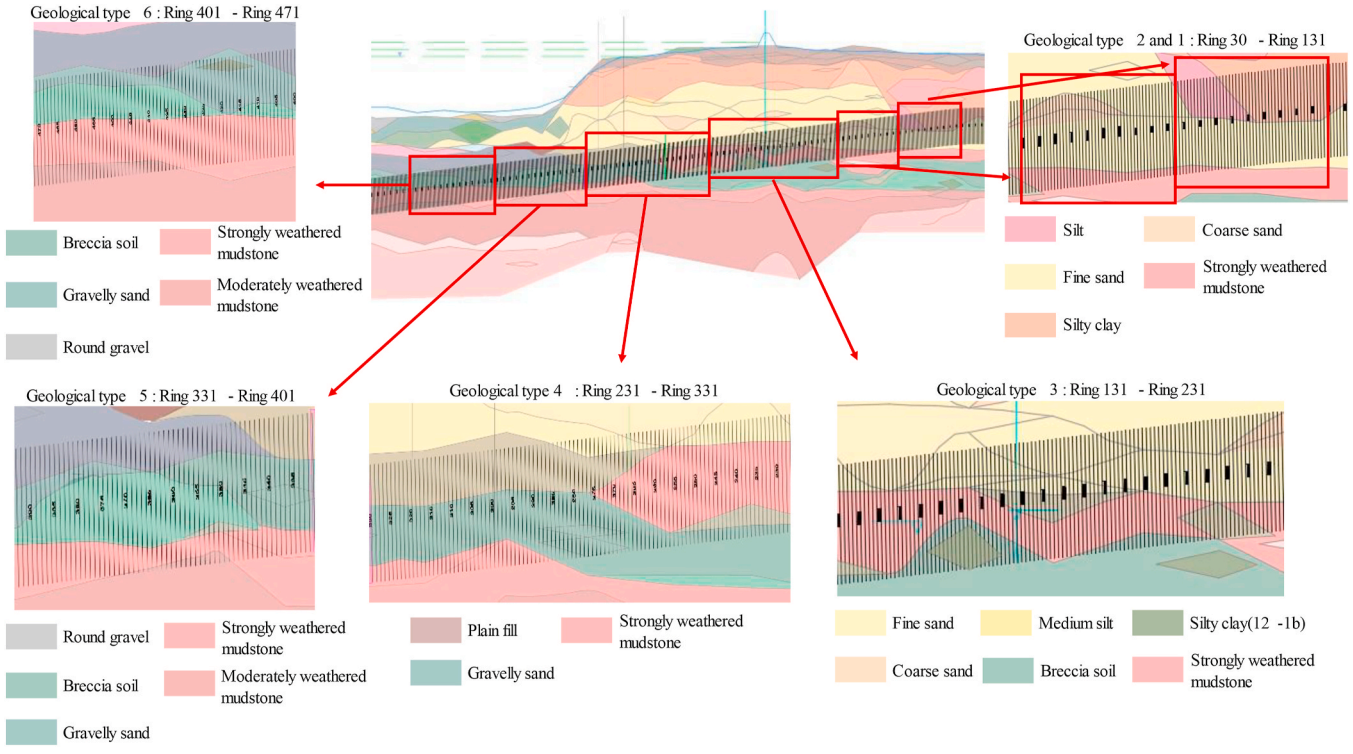


Fig. 10. Geological conditions of the LDSS construction tunnel.

hyperparameters largely determine the capacities of the models and affect their training efficiency. The hyperparameters that must be

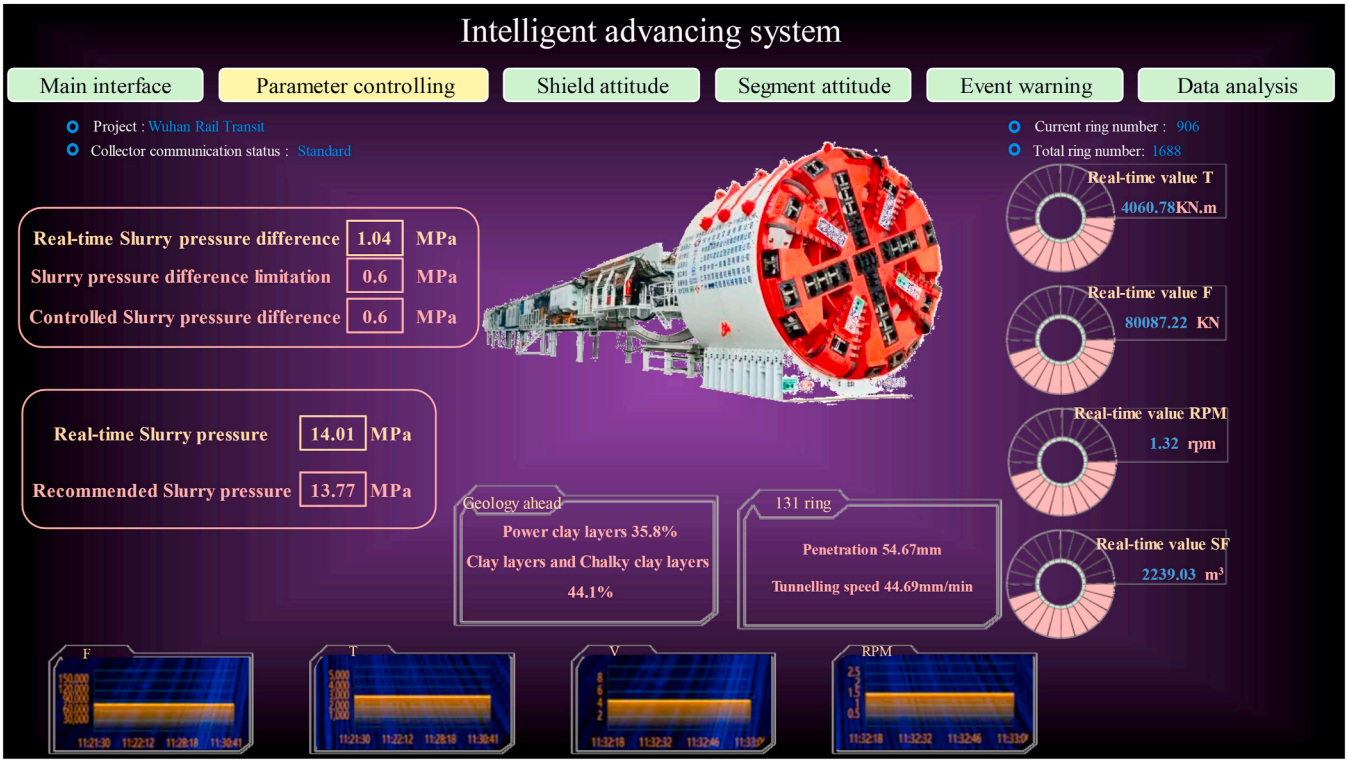


Fig. 11. LDSS construction DT platform interface.

Table 3
Properties parameters of the soils included in each Geological type.

Geological type	Angle of internal friction ϕ (°)			Cohesion c (kN/m^2)			Young's Modulus E (MPa)		
	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean
1	34.5	10.0	23.8	17.0	0.0	9.0	23.8	4.6	14.1
2	34.5	30.0	32.3	32.0	0.0	16.0	43.0	23.8	33.4
3	40.0	6.0	27.9	10.0	0.0	2.5	131.0	101.0	65.1
4	31.0	10.0	23.7	17.0	0.0	5.7	101.0	4.6	49.5
5	40.0	30.0	33.7	32.0	0.0	10.7	133.0	43.0	92.3
6	31.0	21.0	27.3	73.0	0.0	35.0	174.0	43.0	106.0

Table 4
Construction parameter characteristics of Geological type 3.

Parameter	Unit	Std	Mean	Min	Max
x_1	MPa	0.015	0.406	0.359	0.447
x_2	kg·m^{-3}	0.08	1.24	1	1.39
x_3	m^3	155.81	2310.22	1481.37	2980.09
x_4	kg·m^{-3}	0.07	1.32	1.04	1.5
x_5	m^3	170.75	2392.23	840.19	3500
x_6	MPa	0.39	4.11	3.13	6.92
x_7	MPa	5.56	30.62	5.07	59.23
x_8	m^3	12866.06	28898.87	0.3	127900
x_9	mm·min^{-1}	5.39	19.67	0.11	58.87
x_{10}	rpm	0.33	1.4	0.59	2.1
x_{11}	kN·m	965.46	4040.93	250.5	6377.14
x_{12}	kN	10373.67	94813.72	37266.45	130737.6
x_{13}	kN	1199.47	5907.15	4261.47	9553.19
x_{14}	kN	1186.68	6822.59	2399.78	10626.73
x_{15}	kN	5392.72	23019.6	11362.75	34814.54
x_{16}	kN	1502.65	12597.97	8410.73	17203.16
x_{17}	kN	1724.15	11743.03	2474	17448.79
x_{18}	mm	10.41	33.2	0.17	96.99
y_1	MPa	1.01	13.55	11.71	18.32

adjusted for the GCN-BiLSTM model proposed in this paper include the learning rate and GCN-BiLSTM blocks. The MSE is used as the objective function. The number of training iteration is set to 100. The hyper-parameter optimization process and the obtained results are shown in

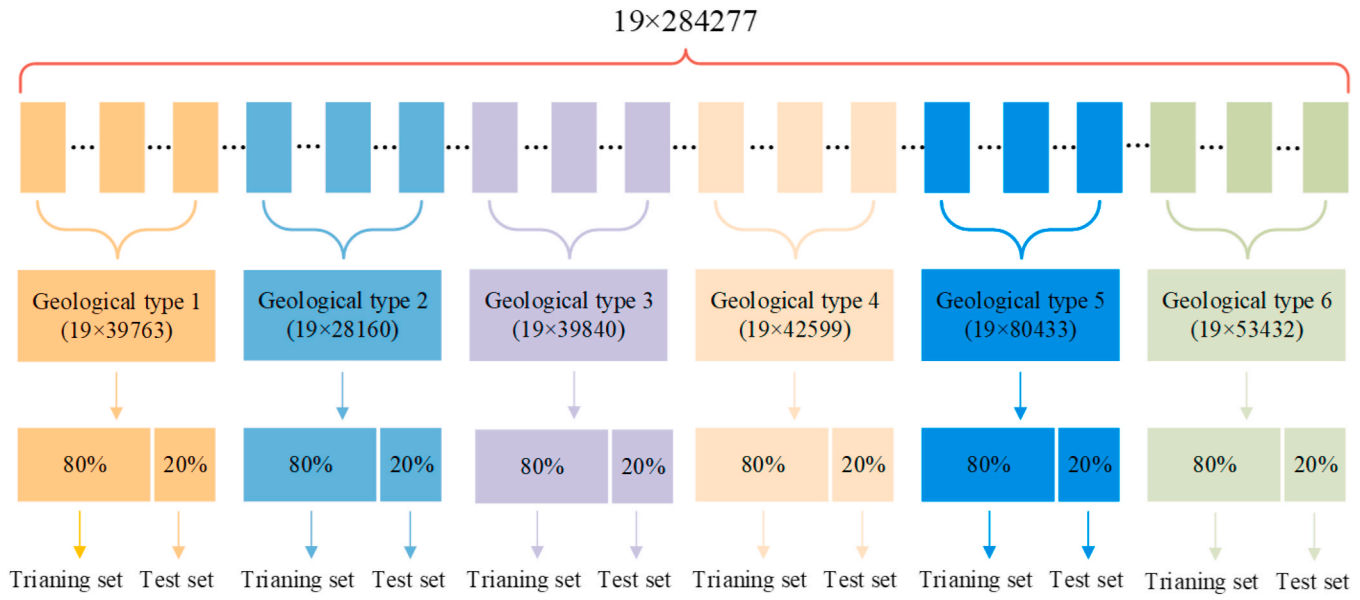


Fig. 12. Schematic diagram of the geological type classification process.

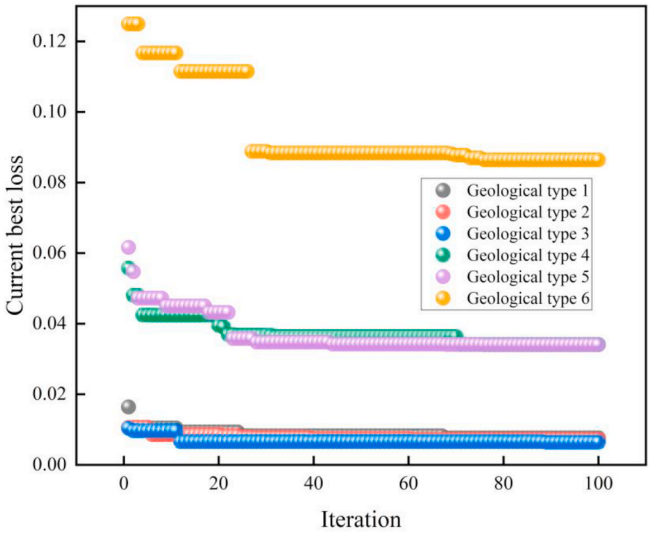


Fig. 13. Iterative process of hyperparametric optimization.

Fig. 13 and Table 5.

Table 5
Hyperparametric optimization results.

Geological type	Learning rate		Blocks	
	Search range	Optimal results	Search range	Optimal results
Geological type 1	[0.001,0.012]	0.009	[1,3]	3
Geological type 2	[0.001,0.012]	0.007	[1,3]	3
Geological type 3	[0.001,0.012]	0.008	[1,3]	3
Geological type 4	[0.001,0.012]	0.008	[1,3]	3
Geological type 5	[0.001,0.012]	0.009	[1,3]	3
Geological type 6	[0.001,0.012]	0.007	[1,3]	3

3.3.2. *Knowledge-data driven method based prediction results*

The number of historical time steps also has an important effect on the prediction accuracy and prediction performance of the model, to assess the validity and accuracy of the prediction results obtained by the proposed model, predictions are made for data with numbers of historical time steps ranging from 1 to 10. For example, based on the use of 3 time steps to predict the SP, the data of the LDSS construction parameters x_1 to x_{18} from moment t-3 to t and the historical data of y_1 from t-3 to t-1 are used as inputs to predict the data of y_1 at moment t. That is, this paper employs a historical window size of 10, a prediction step of 1, and a sliding window size of 1 for time series prediction. BO-GCN-BiLSTM method is built on the basis of BO to obtain hyperparameter combinations for the selected historical time steps. The R^2, RMSE, MAE, and MAPE are used to evaluate the performance of BO-GCN-BiLSTM. Tables 6–11 show the results and training speed of the BO-GCN BiLSTM method for historical time steps t-1 to t-10 in six geological types. Fig. 14 shows the scatter plot of predicted optimal time steps for each Geological type (see Table 12).

(1) The proposed DT platform, based on the knowledge-data driven method, demonstrates excellent predictive performance across six geological types at various time steps. This study predicts SP using time steps t-1 to t-10, and the evaluation metrics for the six geological types are summarized in Tables 6–12.For Geological Type 1, the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.949, 0.061, 0.024, 0.12%. For Geological Type 2,

Table 6
Predicted results and training speed of Geological type 1.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.945	0.058	0.030	0.126%	1.5 epoch/s
t-2	0.950	0.058	0.029	0.115%	1.6 epoch/s
t-3	**0.957**	**0.048**	**0.012**	**0.110%**	**1.9 epoch/s**
t-4	0.944	0.064	0.023	0.129%	1.8 epoch/s
t-5	0.942	0.064	0.026	0.130%	1.7 epoch/s
t-6	0.958	0.062	0.022	0.126%	1.4 epoch/s
t-7	0.950	0.063	0.023	0.129%	1.6 epoch/s
t-8	0.941	0.064	0.029	0.129%	1.3 epoch/s
t-9	0.956	0.063	0.022	0.096%	1.3 epoch/s
t-10	0.946	0.063	0.024	0.122%	1.4 epoch/s
Average	0.949	0.061	0.024	0.12%	1.6 epoch/s

Table 7
Predicted results and training speed of Geological type 2.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.920	0.064	0.034	1.96%	1.7 epoch/s
t-2	0.923	0.055	0.022	1.93%	1.9 epoch/s
t-3	**0.931**	**0.052**	**0.021**	**1.81%**	**1.9 epoch/s**
t-4	0.924	0.053	0.022	1.87%	1.8 epoch/s
t-5	0.923	0.054	0.022	1.96%	1.9 epoch/s
t-6	0.925	0.052	0.025	1.89%	1.4 epoch/s
t-7	0.928	0.050	0.020	1.80%	1.6 epoch/s
t-8	0.924	0.051	0.020	1.85%	1.7 epoch/s
t-9	0.921	0.052	0.021	1.93%	1.7 epoch/s
t-10	0.947	0.050	0.020	1.85%	1.6 epoch/s
Average	0.927	0.053	0.022	1.88%	1.7 epoch/s

Table 8
Predicted results and training speed of Geological type 3.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.923	0.259	0.180	1.27%	1.5 epoch/s
t-2	0.927	0.257	0.176	1.25%	1.9 epoch/s
t-3	**0.942**	**0.248**	**0.169**	**1.24%**	**2.0 epoch/s**
t-4	0.924	0.254	0.172	1.26%	1.6 epoch/s
t-5	0.934	0.242	0.163	1.13%	1.4 epoch/s
t-6	0.924	0.246	0.172	1.22%	1.6 epoch/s
t-7	0.935	0.222	0.170	1.13%	1.5 epoch/s
t-8	0.946	0.221	0.170	1.11%	1.2 epoch/s
t-9	0.947	0.219	0.152	1.10%	1.2 epoch/s
t-10	0.945	0.224	0.171	1.20%	1.6 epoch/s
Average	0.935	0.239	0.169	1.19%	1.5 epoch/s

Table 9
Predicted results and training speed of Geological type 4.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.942	0.194	0.125	0.82%	1.7 epoch/s
t-2	0.946	0.190	0.111	0.80%	1.9 epoch/s
t-3	**0.953**	**0.187**	**0.108**	**0.75%**	**1.9 epoch/s**
t-4	0.942	0.187	0.127	0.84%	1.7 epoch/s
t-5	0.954	0.177	0.106	0.71%	1.5 epoch/s
t-6	0.930	0.187	0.122	0.82%	1.4 epoch/s
t-7	0.952	0.185	0.111	0.77%	1.6 epoch/s
t-8	0.941	0.188	0.114	0.82%	1.3 epoch/s
t-9	0.937	0.189	0.129	0.86%	1.3 epoch/s
t-10	0.947	0.176	0.119	0.74%	1.6 epoch/s
Average	0.944	0.186	0.117	0.79%	1.6 epoch/s

Table 10
Predicted results and training speed of Geological type 5.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.905	0.258	0.112	0.61%	1.4 epoch/s
t-2	0.913	0.257	0.093	0.59%	1.6 epoch/s
t-3	**0.918**	**0.211**	**0.089**	**0.59%**	**1.8 epoch/s**
t-4	0.890	0.254	0.093	0.61%	1.6 epoch/s
t-5	0.909	0.233	0.088	0.55%	1.5 epoch/s
t-6	0.903	0.257	0.088	0.70%	1.4 epoch/s
t-7	0.917	0.239	0.088	0.65%	1.6 epoch/s
t-8	0.900	0.254	0.095	0.66%	1.2 epoch/s
t-9	0.904	0.223	0.090	0.60%	1.3 epoch/s
t-10	0.913	0.220	0.089	0.58%	1.2 epoch/s
Average	0.907	0.241	0.092	0.61%	1.5 epoch/s

the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.927, 0.053, 0.022, 1.88%. For Geological Type 3, the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.935, 0.239, 0.169, 1.19%. For Geological Type 4, the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.944, 0.186, 0.117, 0.79%. For Geological Type 5, the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.907, 0.241, 0.092, 0.61%. For Geological Type 6, the average R^2, RMSE, MAE, and MAPE values across all time steps are 0.895, 0.238, 0.123, 0.88%. The results show that the model achieves high R^2 values and low RMSE, MAE, and MAPE across all six geological types.

Table 11
Predicted results and training speed of Geological type 6.

Time steps	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
t-1	0.882	0.246	0.131	0.99%	1.7 epoch/s
t-2	0.883	0.230	0.121	0.85%	1.7 epoch/s
t-3	**0.908**	**0.225**	**0.116**	**0.78%**	**1.9 epoch/s**
t-4	0.887	0.240	0.120	0.81%	1.6 epoch/s
t-5	0.912	0.239	0.119	0.81%	1.5 epoch/s
t-6	0.878	0.243	0.126	0.94%	1.5 epoch/s
t-7	0.904	0.240	0.125	0.89%	1.7 epoch/s
t-8	0.890	0.243	0.127	0.90%	1.7 epoch/s
t-9	0.907	0.235	0.121	0.89%	1.3 epoch/s
t-10	0.900	0.239	0.123	0.96%	1.3 epoch/s
Average	0.895	0.238	0.123	0.88%	1.6 epoch/s

Table 12
Prediction indicators and training speed of various geological types at time step t-3.

Geological type	SP(y_1)				
	R^2	RMSE	MAE	MAPE	Training speed
Geological type 1	0.957	0.048	0.012	0.11%	1.9 epoch/s
Geological type 2	0.931	0.052	0.021	1.81%	1.9 epoch/s
Geological type 3	0.942	0.248	0.169	1.24%	2.0 epoch/s
Geological type 4	0.953	0.187	0.108	0.75%	1.9 epoch/s
Geological type 5	0.918	0.211	0.089	0.59%	1.8 epoch/s
Geological type 6	0.908	0.225	0.116	0.78%	1.9 epoch/s
Average	0.935	0.162	0.086	0.88%	1.9 epoch/s

These findings indicate that the knowledge-data driven method within the DT platform possesses strong generalization capability and prediction accuracy, making it suitable for adapting to complex and varied geological conditions. As shown in Fig. 15, the predicted and actual values are displayed on the DT platform, which helps LDSS operators assess excavation performance and supports decision-making for construction parameter adjustment. Consequently, it provides reliable technical support for predicting face stability during LDSS construction processes.

(2) Among the various time steps evaluated, the results for time step t-3 are optimal for achieving high prediction accuracy and training speed. As shown in Tables 6–11, across different geological types, t-3 achieves both high predictive accuracy and high training speed. For instance, in Geological Type 1, at t-3, the R^2, RMSE, MAE, MAPE, and training speed are 0.957, 0.048, 0.012, 0.11%, and 1.9 epochs/s, respectively, with R^2 and training speed exceeding the average values of 0.949 and 1.6 epochs/s, and RMSE, MAE, and MAPE below the average values of 0.061, 0.024, and 0.12%. Similarly, in Geological Type 5, at t-3, the R^2, RMSE, MAE, MAPE, and training speed are 0.918, 0.211, 0.089, 0.59%, and 1.8 epochs/s, respectively, with R^2 and training speed above the average values of 0.907 and 1.5 epochs/s, and RMSE, MAE, and MAPE below the average values of 0.241, 0.092, and 0.61%.

While increasing the time step can improve prediction

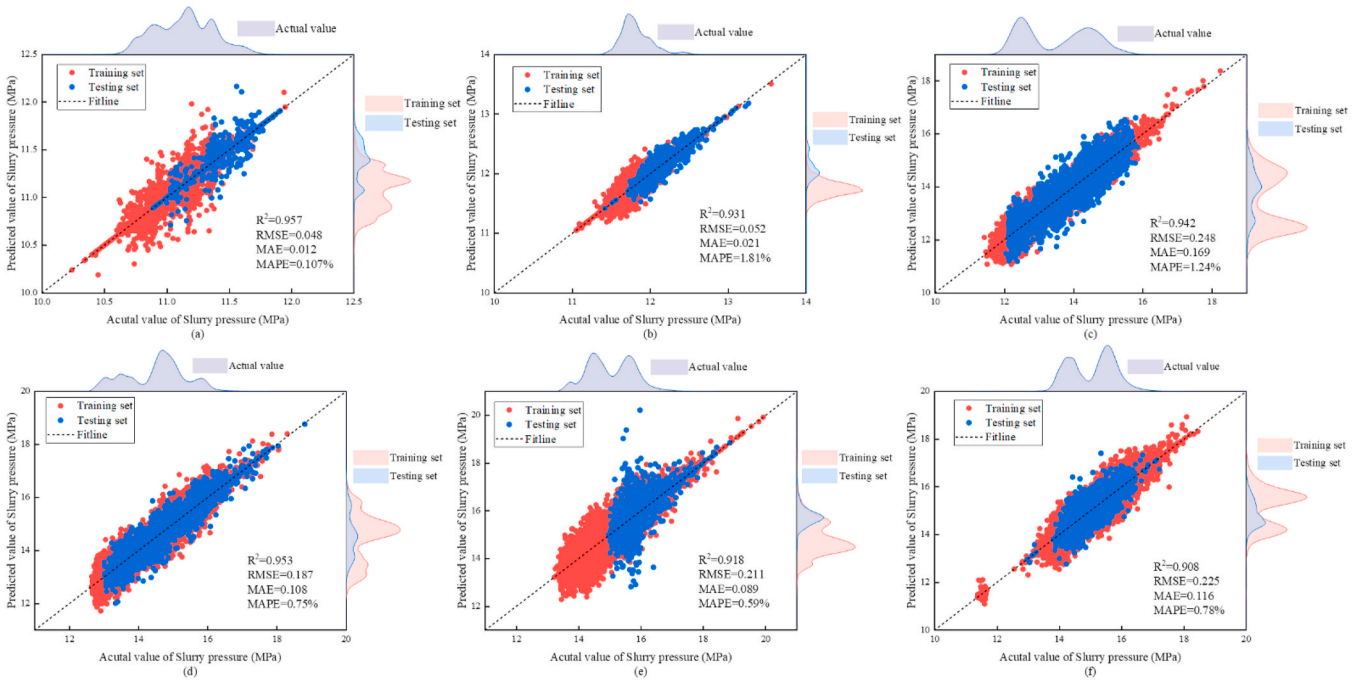


Fig. 14. The best prediction results of SP for different geological types.

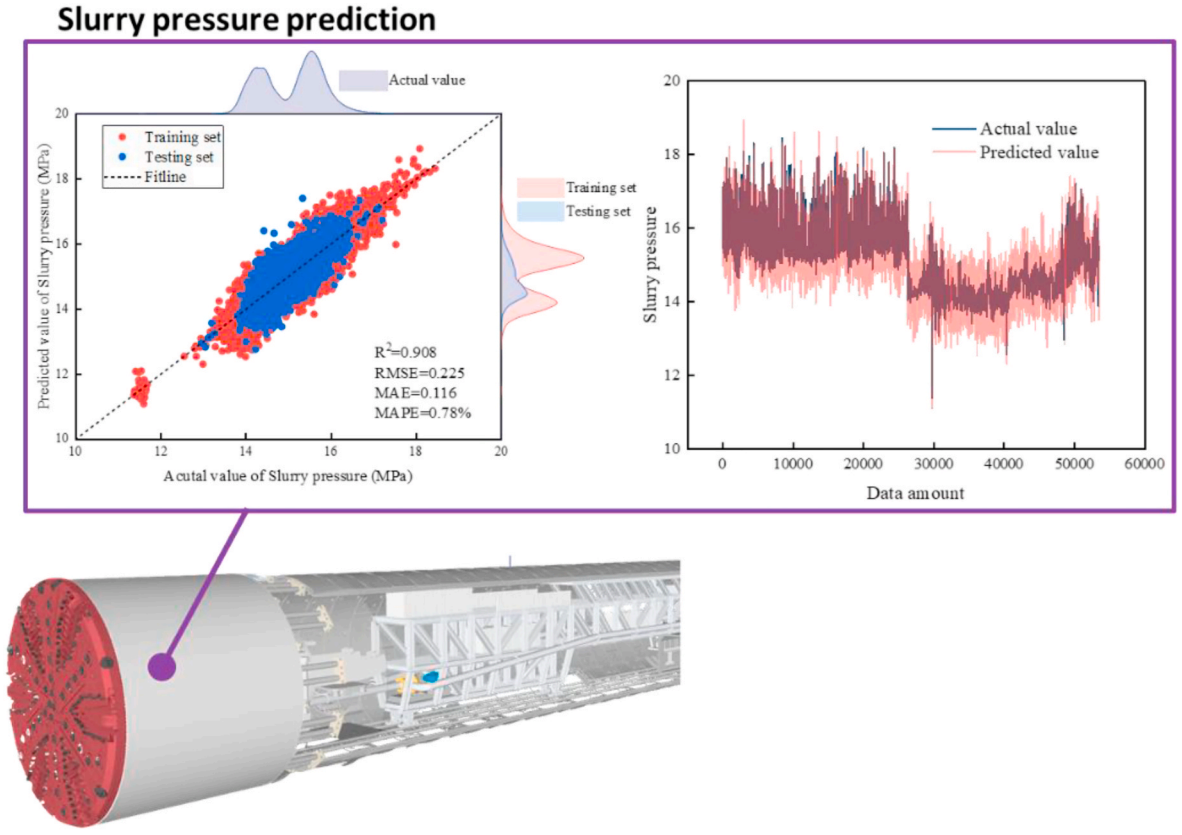


Fig. 15. Virtual demonstration of SP prediction for LDSS operation on the DT platform.

accuracy, it often comes at the cost of reduced training speed. For example, in Geological Type 2, at t-3, the R^2 and training speed are 0.957 and 1.9 epochs/s, whereas at t-10, they are 0.947 and 1.6 epochs/s. In Geological Type 26, at t-3, the R^2 and training speed are 0.908 and 1.9 epochs/s, while at t-5, they are 0.912 and 1.5 epochs/s. Specifically, at time step t-3, the average R^2, RMSE, MAE, and MAPE across six geological types reached 0.937, 0.129, 0.060, and 0.69%, respectively, with an average training speed of 1.9 epochs/s. Therefore, to achieve a balance of high prediction accuracy and high training speed, t-3 is selected as the optimal

time step for prediction.

Fig. 14 presents scatter plots of the prediction results for six Geological types at time step t-3. The predicted points for both the training and test sets are closely aligned with the fit line, consistent with the model's evaluation results. Therefore, the predictions obtained using time step t-3 can provide effective reference data for on-site adjustments of construction parameters.

3.4. Model interpretability analysis

As a powerful tool, SHAP has gained attention for its ability to provide interpretability of model predictions. Through this method, the reality service layer of the DT platform can not only identify the LDSS construction parameters that most significantly impact TFS but also understand how these parameters should be adjusted according to different geological types. This provides deeper insights into ensuring safety during LDSS construction. The SHAP results for each geological type are shown in Figs. 16–18.

As shown in Figs. 16–18, the ranking of construction parameters significantly impacting TFS varies across different geological types. Specifically, the advance pressures A-E (x_{13}-x_{17}) have a notably significant impact on TFS in Geological Types 5 and 6. Particularly in Geological Type 5, construction parameters x_{16} and x_{13} both rank within the top five most influential construction parameters affecting TFS. Conversely, in Geological Types 1 and 2, the influence of these construction parameters is relatively minor, with lower rankings. Moreover, the in and out slurry pressure difference (x_6) exhibits higher influence rankings in Geological Types 1 and 3. However, in Geological Types 2, 4, 5, and 6, x_6 is identified as the most significant construction parameters impacting TFS. Meanwhile, some construction parameters maintain consistent rankings across all six geological types; for instance, Air cushion chamber pressure (x_1) consistently ranks within the top three in terms of its influence on TFS across all geological types, underscoring its consistent importance.

Although SHAP can identify important construction parameters affecting TFS under different geological types, it is essential to distinguish between active and passive parameters in LDSS construction. Active parameters can be directly adjusted by operators, while passive parameters, though identified as important, cannot be controlled. Additionally, some active parameters may rank lower but still require attention during operations. Therefore, based on SHAP and expert knowledge, the final selection of the top five parameters influencing TFS across various geological types are: x_1, x_7, x_9, x_{10}, and x_{11}. The above analysis results also can be verified in similar studies in the past (Chen et al., 2013; Shang et al., 2023; Weng et al., 2020).

In this study, the interpretability of the BO-GCN-BiLSTM model is enhanced through the adoption of SHAP, revealing the complexity and geological dependency of how LDSS construction parameters influence TFS. This provides guidance for identifying key construction parameters affecting the entire well section. Furthermore, SHAP-based interpretation of the data-driven method establishes a theoretical foundation for adaptive control strategies in LDSS operations under varying geological conditions. During construction, LDSS operators can leverage SHAP analysis results from the DT platform combined with integrated expert knowledge to adjust TFS control strategies specifically for different geological types, thereby achieving an adaptive closed-loop TFS control encompassing interpretation, decision-making, and execution.

4. Discussion

4.1. Model generalizability study

Validating the effectiveness of the proposed method is an essential task. The prediction results obtained for the 6 Geological types are compared with those of the existing methods. The methods selected for comparison from previous studies include BO-GCN-BiLSTM, BO-Transformer, BO-GCN-LSTM, BO-BiLSTM, BO-LSTM, BO-GCN. During the prediction process, the hyperparameter optimization method uses BO, and the evaluation indices are R^2 and RMSE. Among them, the property parameters of each geological type are shown in Table 3. The data partitioning of each geological type is shown in Fig. 12 and the

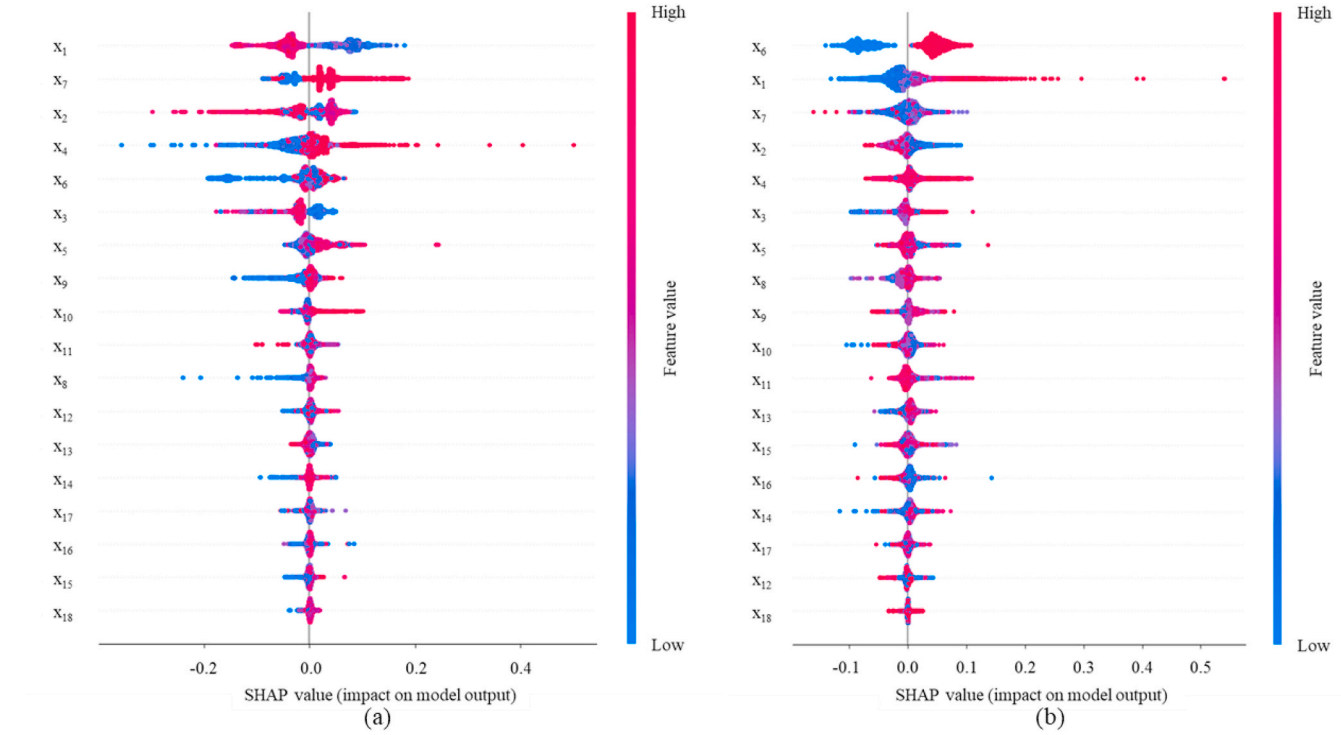


Fig. 16. SHAP results for Geological type 1 and Geological type 2.

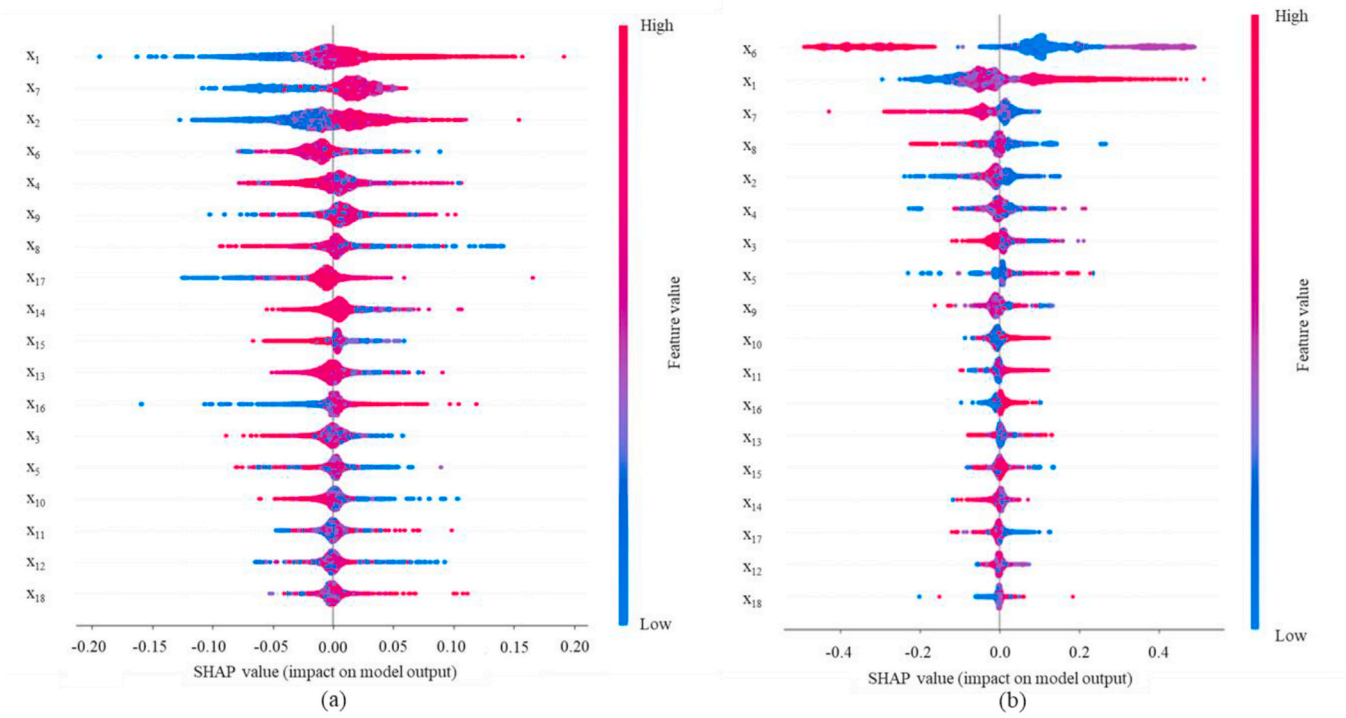


Fig. 17. SHAP results for Geological type 3 and Geological type 4.

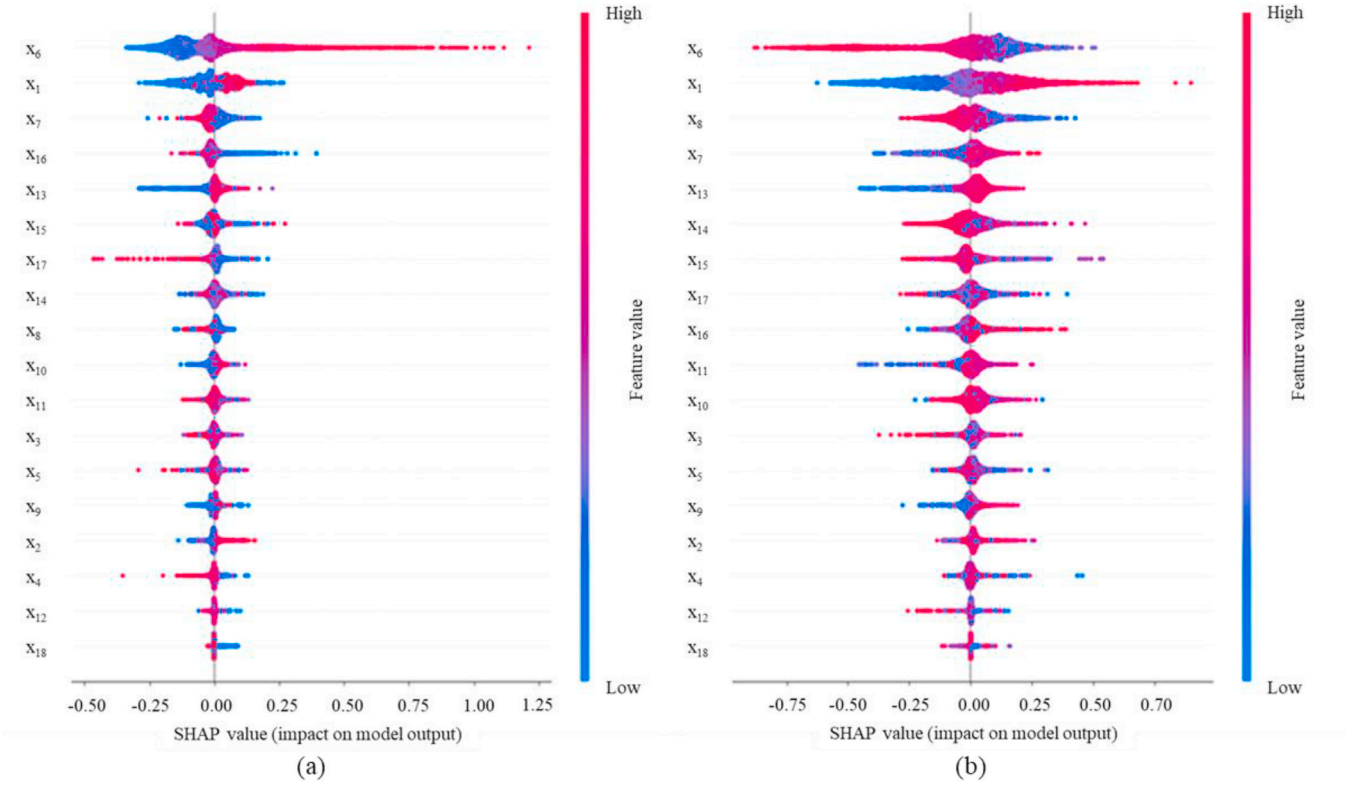


Fig. 18. SHAP results for Geological type 5 and Geological type 6.

prediction results and computational efficiency of each method are provided in Table 13, Table 14 and Fig. 19.

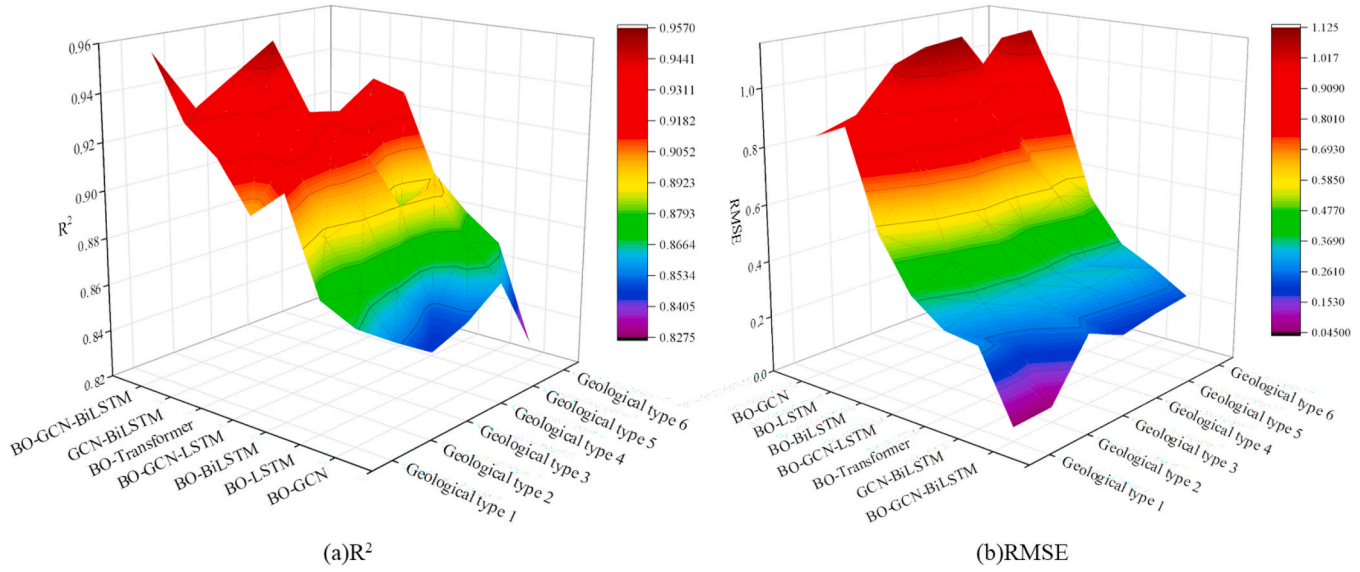
Upon comparison of the prediction outcomes across six geological types, the BO-GCN-BiLSTM demonstrates superior predictive performance. As shown in Fig. 19, for the six geological types, the proposed method predicts the SP with an average R^2 of 0.935 and an average RMSE of 0.162. Compared with those of the other six methods, BO-GCN-BiLSTM achieves the highest average R^2 and lowest average RMSE. the prediction performance of GCN-BiLSTM, is slightly lower than that of BO-GCN-BiLSTM, achieving average R^2 of 0.929 and average RMSE of

Table 13
Prediction indicators for evaluating various methods on different geological types.

Geological types	BO-GCN-BiLSTM		GCN-BiLSTM		BO-Transformer		BO-GCN-LSTM		BO-BiLSTM		BO-LSTM		BO-GCN	
	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
1	0.957	0.048	0.931	0.282	0.920	0.298	0.900	0.383	0.912	0.563	0.874	0.902	0.867	0.841
2	0.931	0.052	0.932	0.230	0.925	0.337	0.922	0.451	0.889	0.767	0.871	0.909	0.855	0.883
3	0.942	0.248	0.939	0.301	0.923	0.351	0.914	0.537	0.894	0.877	0.865	1.107	0.844	1.031
4	0.953	0.187	0.923	0.323	0.914	0.344	0.911	0.489	0.892	0.862	0.850	1.121	0.850	0.921
5	0.918	0.211	0.921	0.344	0.931	0.323	0.880	0.554	0.897	0.699	0.875	0.932	0.861	0.901
6	0.908	0.225	0.932	0.299	0.928	0.360	0.894	0.513	0.880	0.869	0.869	1.099	0.828	1.054
Average	0.935	0.162	0.929	0.296	0.924	0.336	0.904	0.488	0.894	0.773	0.867	1.012	0.851	0.939

Table 14
Comparison of computational efficiency of various methods under different geological types.

Geological types	BO-GCN-BiLSTM	GCN-BiLSTM	BO-Transformer	BO-GCN-LSTM	BO-BiLSTM	BO-LSTM	BO-GCN
	(epoch/s)	(epoch/s)	(epoch/s)	(epoch/s)	(epoch/s)	(epoch/s)	(epoch/s)
1	1.9	1.7	1.6	2.0	1.9	2.3	2.5
2	1.9	1.9	1.5	1.7	2.2	2.2	2.1
3	2.0	1.7	1.7	1.9	2.0	2.6	2.1
4	1.9	2.1	1.9	1.8	2.1	2.0	2.2
5	1.8	1.7	1.5	1.9	2.4	2.1	2.6
6	1.9	1.9	1.6	1.9	2.1	2.3	2.7
Average	1.9	1.8	1.6	1.9	2.1	2.3	2.4



(a)R^2 (b)RMSE

Fig. 19. Average results of SP prediction by each method for the six geological types.

0.296; The prediction performance of BO-GCN is relatively low, with an average R^2 of 0.851 and average RMSE of 0.939.

As shown in Table 14, BO-GCN-BiLSTM does not achieve the highest computational efficiency, with an average of 1.9 epoch/s. BO-BiLSTM, BO-LSTM, and BO-GCN exhibit higher computational efficiencies of 2.1 epoch/s, 2.3 epoch/s, and 2.4 epoch/s, respectively, although the differences are not substantial. However, these three models demonstrate inferior prediction accuracy compared to BO-GCN-BiLSTM. GCN-BiLSTM and BO-GCN-LSTM show computational efficiencies comparable to BO-GCN-BiLSTM, yet they still underperform in terms of prediction accuracy. BO-Transformer exhibits the lowest computational efficiency. In summary, considering both prediction accuracy and computational efficiency, the proposed BO-GCN-BiLSTM demonstrates the best performance for TFS prediction.

The BO-GCN-BiLSTM outperforms other state-of-the-art methods in SP prediction for LDSS tunneling due to its synergistic integration of three key components: (1) GCN's capacity to explicitly model spatial coupling relationships among critical parameters (Dun et al., 2022), (2) the BiLSTM's ability to comprehensively capture temporal dependencies through forward and backward memory mechanisms (Wei et al., 2025), and (3) BO's probabilistic surrogate-based strategy for efficiently identifying globally optimal hyperparameters with fewer iterations (Chakrabarty et al., 2023). This integrated architecture achieves higher prediction accuracy while maintaining stable generalization error across six distinct geological data distributions—attributable to the topological robustness of GCN and the bidirectional temporal modeling of BiLSTM—thereby significantly reducing both retraining overhead and inference time.

Therefore, the proposed method not only provides robust support for intelligent decision-making in LDSS tunneling but also offers a precise and effective artificial intelligence–assisted approach for risk prediction and management in future tunneling projects under complex geological conditions. Its knowledge-data driven mechanism can significantly enhance safety margins and adaptive control capabilities in

underground operations. Moreover, the framework can be rapidly transferred to other underground engineering scenarios—such as metro systems, utility tunnels, and deep mining operations—delivering an advanced solution for intelligent construction and safe operation of critical infrastructure in geologically challenging environments.

4.2. Knowledge-data driven method based TFS control

The proposed DT platform, based on a knowledge-data driven method, enables accurate prediction of TFS and, combined with SHAP, makes adaptive TFS control feasible. Within the DT platform, the reality service layer identifies the key construction parameters influencing TFS through SHAP analysis, providing theoretical guidance for LDSS operators. This assists LDSS operators in timely adjusting relatively optimal control strategies during different construction phases and under varying geological conditions, thereby achieving SHAP-enhanced, DT platform-based adaptive TFS control.

For SP, in LDSS, the shield tunneling process needs to control the amount of slag in the cutting machine used to cut the original soil into slurry soil and the amount of slurry in the slurry soil after mixing in the slurry inlet pipe. The volume of mixed mud in the mud soil discharged from the mud discharge pipe is equal to ensure that the mud in the mud soil provides stable SP. The calculation formula for the adjusted SP is shown in the empirical formula Eq. (16) (Zhou zhong et al., 2022). Therefore, to control the TFS during LDSS construction, the difference between the current SP and the adjusted SP need to be as small as possible; thus, the TFS of the LDSS is in a safe state. Existing research and field application results have validated the effectiveness and applicability of this formula (Ge, 2025; Ren et al., 2025; Shi et al., 2025b; Xu and Huang, 2024).

$$p_{k+1} = p_k + \frac{E(p)}{4\rho_2 V_c}\left(\pi D^2 v_a \rho' + 4q_1\rho_1 - 4q_2\rho_2\right)\Delta T \tag{16}$$

where p_k and p_{k+1} are the current SP and the adjusted SP, respectively; $E(p)$ is the bulk modulus of the mixed slurry in the mud–water soil; V_c is the volume of the mud–water soil; D is the diameter of the cutter disc; v_a is the propulsion speed; ρ' is the in situ soil density; ρ_1 and ρ_2 are the density of the mud at the feed pipe and the density of the mixed slurry at the discharge pipe, respectively; q_1 and q_2 are the feed flow and the slurry discharge flow rate; and ΔT is the period.

Considering the differences between active and passive construction parameters in LDSS construction, the DT platform employs a knowledge-data driven method that combines expert knowledge with data-driven method to distinguish these parameters. For TFS control, the following construction parameters were selected: x_1, x_7, x_9, x_{10}, x_{11}, x_{17}, x_{14}, x_{15}, x_{13} and x_{16}. Using the empirical formula provided in Equation (15), the warning value range was calculated as [−0.6, 0.6] MPa. Based on 1000 data points from Geological Type 3, continuous adjustments of the aforementioned construction parameters are made until SP falls within the warning value range, after which the next round of control adjustments is initiated.

The adjustment ranges for each construction parameter are shown in Table 15, and the TFS control interactive workflow is illustrated in Fig. 20. As illustrated in Fig. 20, key construction parameters and their control ranges are transmitted via data streams to the DT platform, where the TFS control process is executed based on the virtual entity. The resulting control outputs from the DT platform are fed back to the LDSS operator, who adjusts the critical parameters accordingly. Thus, the interaction between the DT platform and the LDSS operator facilitates adaptive TFS control.

Table 15
Adjustment range for each construction parameter.

Construction parameter	Value range
Air cushion chamber pressure (MPa, x_1)	0.3–0.5
Slurry injection pressure (MPa, x_7)	10~50
Tunnelling speed (mm·min^{-1}, x_9)	20~60
Cutter head speed (rpm, x_{10})	0.6–2
Cutter head torque (kN·m, x_{11})	500~6000
Advance pressure E (kN, x_{17})	4300~10000
Advance pressure B (kN, x_{14})	3000~10000
Advance pressure C (kN, x_{15})	15000~30000
Advance pressure A (kN, x_{13})	5000~9000
Advance pressure D (kN, x_{16})	10000~15000

5. Conclusions and future work

This study establishes a DT platform that enables interaction between the physical entity of LDSS and its virtual model, facilitating prediction and control of TFS during LDSS construction. The platform's Virtual Twin Layer constructs the virtual entity of LDSS, collecting physical data from the LDSS monitoring system and integrating expert knowledge. The Data-driven Module, which integrates BO, GCN, and BiLSTM, provides technical support for achieving high-precision TFS prediction and control. Through the establishment of the Reality Service Layer, the platform implements TFS prediction and control based on a knowledge-data driven method. The developed DT platform serves as an intelligent tool to assist LDSS tunnel construction and enhance safety across different geological types. The Wuhan Metro project is used as a case study to validate the effectiveness of the established DT platform and the knowledge-data driven method. The results indicate:

(1) The BO-GCN-BiLSTM method proposed in this paper can accurately predict TFS during LDSS construction. Specifically, the average R^2, RMSE, MAE, and MAPE across six geological types reached 0.937, 0.129, 0.060, and 0.69%, respectively, with an average training speed of 1.9 epochs/s. Furthermore, compared with the other methods, such as GCN-BiLSTM, BO-Transformer, BO-GCN-LSTM, BO-BiLSTM, BO-LSTM, BO-GCN, the BO-GCN-BiLSTM model demonstrated superior performance in predicting TFS across six different geological types, with an average R^2 of 0.935 and an RMSE of 0.162.
(2) Expert utilize SHAP and their own expert knowledge, key influencing construction parameters were specified to improve the interpretability of the knowledge-data driven method and rank the significant impacts; here, air cushion chamber pressure (x_1), slurry injection pressure (x_7), tunneling speed (x_9), cutter head speed (x_{10}) and cutter head torque (x_{11}) are the top five construction parameters. These parameters are the key construction parameters for controlling the TFS diuring LDSS construction; therefore, priority should be given to adjusting these construction parameters in the control of the TFS diuring LDSS construction.
(3) By integrating the knowledge-data driven method with DT, a DT platform has been developed for data collection, TFS prediction and control during LDSS construction. The system features an intuitive interface for the intelligent propulsion system of the LDSS, which visually displays data on the excavation status and construction parameters of the LDSS. This enables the DT platform to adjust key construction parameters for effective control of the TFS diuring LDSS construction.

Although the BO-GCN-BiLSTM method implemented within the DT platform demonstrates strong performance in capturing spatiotemporal dependencies among construction parameters in LDSS, it exhibits notable limitations in handling geological complexity and lacks physical interpretability. During shield tunneling, the intricacy and dynamic evolution of geological conditions constitute critical factors influencing face stability. While BO-GCN-BiLSTM is capable of processing complex spatiotemporal data, its adaptability may be insufficient when confronted with abrupt geological transitions or previously unseen geological formations. This limitation stems from the model's reliance on historical data for learning, which hinders its ability to generalize

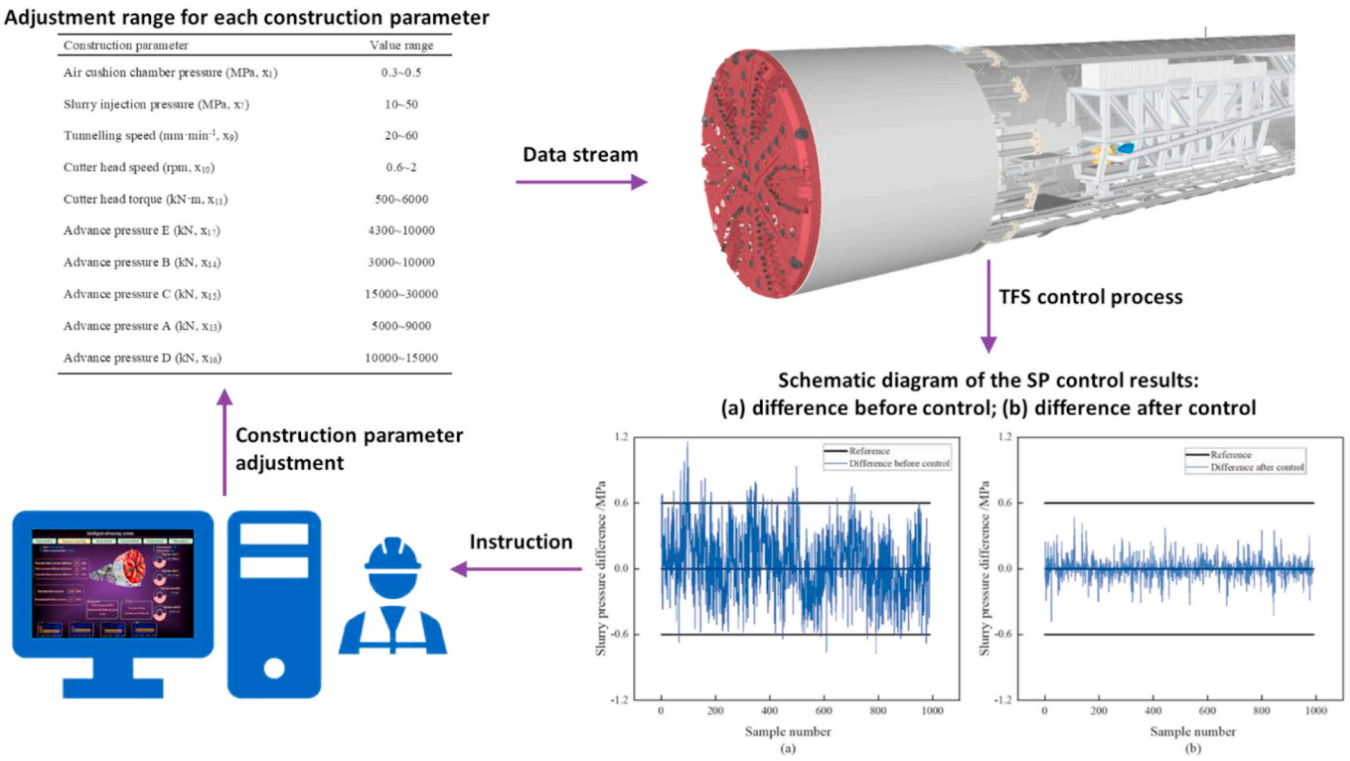


Fig. 20. Interactive workflow demonstration of TFS control during LDSS operation.

effectively to novel geological scenarios. Furthermore, although the DT platform can enhance model interpretability and refine key parameter selection through the integration of expert knowledge—thereby improving reliability—this process remains inherently susceptible to subjective judgment and heavily dependent on the practitioner's experience. Future research should aim to embed geological priors and governing physical equations directly into the network architecture, enabling the model to rapidly self-correct using domain-specific knowledge when encountering sudden geological changes. Such an approach would not only strengthen the model's adaptability to highly complex and variable ground conditions but also improve its interpretability from the perspective of underlying physical laws.

CRediT authorship contribution statement

Xianguo Wu: Writing – original draft, Funding acquisition. **Yu Lei:** Formal analysis, Data curation. **Feiming Su:** Methodology, Investigation. **Tiejun Li:** Supervision, Resources. **Yang Liu:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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